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The Effect of Maternal Education on Fertility and Infant Health: Evidence From School Entry Policies Using Exact Date of Birth*

Justin McCrary
University of Michigan

Heather Royer
University of Michigan

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Abstract

This paper uses age-at-school-entry policies to identify the effect of maternal education on fertility and infant health. We focus on sharp contrasts in schooling, fertility, and infant health between mothers born just before and after the school entry date. School entry policies affect maternal education and the quality of a mother's mate but have small and insignificant effects on fertility and infant health. We argue that school entry policies manipulate primarily the education of young women at risk of dropping out of school. For these women, education may play a limited role in fertility and child investment.

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Maternal education is widely held to be a key determinant of fertility and infant health. From a theoretical perspective, several causal channels have been emphasized. First, education may raise a woman's permanent income directly through the earnings channel, tilting her optimal fertility choices toward fewer offspring of higher quality (Mincer 1963, Becker 1960, Becker and Lewis 1973). Second, positive assortative mating leads highly educated women to pair with highly educated men (Behrman and Rosenzweig 2002), further raising household permanent income and altering optimal fertility choices. Third, education may directly improve an individual's knowledge of, as well as ability to process information regarding, fertility options and healthy pregnancy behaviors (Grossman 1972).

On the empirical side, an extensive literature documents associations between maternal education and fertility and infant health that are consistent with this reasoning (Strauss and Thomas 1995). However, whether these associations represent causal relationships has been the subject of a spirited debate. Early quasi-experimental research on differences in education between sisters who become mothers points toward smaller effects than the cross-sectional relationship, suggesting an important role for selection (see, for example, Wolfe and Behrman 1987). On the other hand, more recent quasi-experimental research focused on primary school construction programs in Taiwan (Chou, Liu, Grossman and Joyce 2003) and Indonesia (Breierova and Duflo 2004), and on college openings in the United States (Currie and Moretti 2003), suggests that there is a causal effect, and in the latter case suggests that simple observational comparisons may actually understate the true causal effect. Recent unpublished quasi-experimental papers by Black, Devereux and Salvanes (2004) and Leon (2004) similarly support the notion that the causal effect is as large as the partial correlation.

In this paper, we present new evidence on the effect of maternal education on fertility and infant health in the United States using school entry policies as an instrument for maternal education. In particular, we exploit the fact that the year in which a child starts school is a discontinuous function of exact date of birth. For example, in California and Texas, our two study states, children must be 5 years old on December 1st (California) or September 1st (Texas) in the year in which they begin kindergarten. As a consequence of these policies, children born within one or two days of one another enter school at different ages and have different levels of education throughout school enrollment.[1] Because individuals born near in time are likely similar, any differences in education at motherhood for individuals born near these entry dates are arguably exogenous. The crux of our identification strategy is to compare fertility and infant health outcomes for mothers born just prior to, and just subsequent to, the school entry date, and to relate the magnitude of these differences to the maternal education discontinuity.

Using large samples of birth records for the two states, we reach the following four key

[1]This assertion assumes grade progression is unaffected by school start age. Matsudaira and McCrary (2004) provide evidence that the discontinuity in grade-in-school is equal for third through eighth grade.

conclusions:

1. school entry policies have a significant impact on schooling at motherhood: one-fourth of young Texas mothers born after the school entry date have a year less education than they otherwise would, had they been born before the entry date. For California, our estimate is one-seventh. These education discontinuities are estimated precisely, with a t-ratio over 20 for Texas and over 8 for California.

2. Maternal education does not significantly impact fertility: women born just before and after the school entry date are equally likely to become mothers, and these women give birth at similar ages.

3. Maternal education does improve mating market outcomes: women born just after the entry date have younger and lesser educated mates than women born just before.

4. Maternal education does not significantly impact observable inputs to infant health or infant health directly: women born just before and after the entry date give birth to children of similar health endowments, as proxied by birth weight, prematurity, and rate of infant mortality, and have similar prenatal behaviors, as measured by rates of smoking and prenatal care.

Given the third effect on mate quality, our infant health results are particularly surprising, as the combined effect of the direct impact of mother's schooling and the indirect impact of her mate's quality is zero in terms of infant health. Assortative mating effects would have been expected to lead to important infant health differences, even if maternal education were unimportant in isolation.

This methodology builds on the original quarter of birth approach of Angrist and Krueger (1991).[2] First, it allows us to overcome the criticisms of Bound, Jaeger and Baker (1995) regarding seasonality of birth. Second, it leads to a more precise estimate of the relationship between within-year birth timing and educational attainment, allowing us to sidestep statistical issues pertaining to weakly correlated instrumental variables.[3]

A practical difficulty with this methodology is that few public-use data sets have exact date of birth. We take advantage of the fact that mother's date of birth (i.e., not her delivery date) was added to the standard U.S. birth certificate in 1989. Because this information is not available in public-use birth certificate data sets, we use restricted-access data from California and Texas for all women giving birth in these states between 1989 and the present. The data contain measures of education and maternal date of birth as well as measures of infant health, pregnancy behaviors (e.g., smoking and drinking), and paternal characteristics.

[2]The Angrist and Krueger approach is a "second best" strategy the authors were compelled to adopt because of the limitations of the public-use micro-sample files.

[3]Moreira (2002) describes tests of correct size and provides references to the literature.

Because of the statistical power furnished by these data, our analysis focuses narrowly around the school entry date, a challenge for earlier analyses, for which either exact date of birth or large sample sizes were wanting (Angrist and Krueger 1991, Cascio and Lewis 2005).

The crucial assumption underlying this approach is that for dates near the school entry cutoff date, an individual's date of birth is random. *A priori* this assumption appears to be reasonable as it seems unlikely that parents strategically plan the exact date of birth of their child. Moreover, this assumption is testable. In particular, women born just before and just after the entry dates should be similar in terms of pre-determined, observable characteristics. We find that they are.

Our main results are based on women who are 23 years old or younger at the time of childbirth.[4] A non-trivial share of these women may be still enrolled in school. These women are unlikely to have benefitted yet from the indirect effects of their education - most importantly, the income returns of their education. As such, our estimates may be primarily isolating the direct effect of education via learning and the ability to process information. On the other hand, young women may accurately forecast their completed education and household resources and choose their fertility and child investment accordingly, in which case the effect for those enrolled in school may reflect both the indirect and direct effects of their education. One way to disentangle these competing interpretations is to stratify the sample by maternal age. The youngest mothers are likely those whose schooling was interrupted and the oldest mothers are likely those who have completed their education. Since the estimates by maternal age show no consistent patterns, our estimates may suggest that both the direct and indirect effects of maternal education are negligible.

The conclusions above are subject to a caveat regarding age-in-grade effects. Our comparison of mothers born just prior and just subsequent to the school entry date implicitly is a contrast of women who are youngest in their academic cohort with those women who are oldest in their grade. Consequently, a competing interpretation to those outlined above is that education *does* in fact affect a woman's fertility choices and investments in healthy pregnancy behaviors, but that the effects are offset by an old-for-grade effect. We discuss this possibility in detail in Section IV.F. We conclude that age-in-grade effects are not a likely explanation for the observed patterns in fertility and infant health by day of birth.

Even if both education and old-for-grade effects are operative, our empirical results continue to have an interpretation as the program evaluation of postponing schooling as it pertains to fertility and infant health. This policy evaluation is relevant both to the pri-

[4] As to be discussed later, we base our sample on these women because of the strength of the first-stage relationship between the school entry policies and education for this sub-population. This relationship is strongest for these women for two reasons. First, some of these women may still be enrolled in school. The effect of the school entry policies on education should weaken as individuals complete their education. Second, the policies manipulate mainly the number of years of high school education. Women who delay childbearing tend to have higher levels of education and thus, are less affected by the policies.

vate decisions of parents contemplating when their children should start school, as well as to the current debate regarding the appropriate entry date. Several states have recently moved, or are currently debating moving, these dates from late in the year to the early Fall (Aizenman 2002), with the stated rationale of raising the age of the average kindergartner. While starting children at older ages may help them cope with the demands of an increasingly rigorous kindergarten curriculum, our results suggest that, for some girls, doing so makes it more likely that pregnancy will interrupt school progression at an earlier grade. Our results suggest this will lead to reduced completed schooling and mates of lower education and earnings ability.

The remainder of the paper is organized as follows. In Section I we describe in more detail the mechanisms by which education could affect fertility and infant health and briefly summarize the existing literature on the topic. In Section II, we turn to a discussion of our identification strategy and discuss estimating equations. After describing the data we use in Section III, we present the results of our estimation in Section IV. Section V compares our estimates with those from the existing literature, and Section VI concludes.

I. Why Should Education Matter?

In broad terms, education may affect a woman's fertility and child-investment choices through either income or learning. Education increases a woman's income stream through both the labor market and the mating market, the latter through assortative mating. In addition to the income channel, education may improve a woman's stock of knowledge regarding contraceptive technologies or healthy pregnancy behaviors, either because it augments her knowledge directly (i.e., educational curricula are important), or because it improves her ability to absorb and process information generally. We next describe each of these mechanisms in turn.

A key reason to suspect that education affects fertility and infant health is the well-documented effect of education on labor earnings. In a comprehensive survey, Card (1999) concludes that by the 1990s, the causal effect of an additional year of education on annual earnings for both women and men in the United States was around 15%.[5,6] The notion that an exogenous increase in a woman's income may lead to reduced fertility is present in the earliest treatments of the neoclassical model of fertility (Mincer 1963, Willis 1973). In these models, households do not value children *per se*, but what Willis terms "child services"—the

[5]See Table 1 and the summary at p. 1855. These estimates take the parameter to be defined as the effect of one year of education, holding constant potential experience. This is mechanically larger than the effect of one year of education, holding constant age (i.e., there is a necessary tradeoff between education and experience). Note also that the return to education has been rising in the U.S. in recent years.

[6]Taking the return to education to be constant over the life cycle, this implies that an exogenous additional year of schooling raises by 15% an individual's present discounted value of expected income, or Friedman's (1957) concept of permanent income. Permanent income seems closest to the notion of income used in most of the early static models of fertility. See, for example, the discussion of Willis (1973, p. S22, S48, S52).

product of the number of children and the average quality of those children. A key idea is that production of child services is time-intensive relative to other activities for the woman. As the value of a woman's time rises, she generally substitutes away from consumption that is highly time-intensive (Becker 1965) and hence desires fewer or perhaps no children. These predicted effects of education on fertility map naturally into predicted effects on child quality. Assuming child services are a normal good, falling fertility in response to rising income requires that child quality be an increasing function of income. Cross-price effects such as these were first emphasized by Becker and Lewis (1973).[7]

These predictions are further sharpened by positive assortative mating, or the tendency for women of relatively higher education to pair with men of similar education (Behrman and Rosenzweig 2002). Under this type of stratification, an exogenous increase in a woman's education leads her to find a mate of higher education, further increasing household permanent income through a multiplier effect.

In addition to the income channel, the literature has stressed the role of education in augmenting an individual's stock of health knowledge.[8] With respect to fertility, Rosenzweig and Schultz (1989) provide evidence that a woman's education explains ability to effectively use contraception. With respect to infant health, Thomas, Strauss and Henriques (1991) show that education predicts a woman's ability in regards to, or perhaps interest in, information acquisition and processing.[9] One of the most frequently-cited examples is smoking (Currie and Moretti 2003). Through anti-smoking campaigns in schools or health class, children could learn about the dangers of smoking and be discouraged from adopting the habit. Glewwe (1999) argues that the most important mechanism for knowledge gain is not directly via curricula; rather the skills obtained in school facilitate the acquisition of health knowledge. Grossman (1972) formalizes these ideas by viewing education as a productivity shifter in the household production function for health.

Since education can affect infant health through several different channels and the intensity of these channels may not be the same for all levels of education nor for all subpopulations, the effect of education on infant health may differ across studies. For example, Currie and Moretti (2003) use college openings to study the effect of maternal education on infant health. The women whose schooling attainment at motherhood is affected by college openings are those women with a high level of education generally. As we show below, our study focuses on the causal role of education for women with a low level of education generally. Educational levels that appear to be affected in our study are in the range of eighth grade to twelfth grade, with a muted effect on the first two years of college. It is interesting to recall Willis's (1973, p. S52) conjecture that education reduces fertility especially "in populations

[7]For an early empirical examination of this tradeoff using twins, see Rosenzweig and Wolpin (1980).
[8]See, for example, Willis (1973, p. S51)
[9]For a contrary view in regards to adult health, see Kenkel (1991).

or subpopulations in which [the] wife's education levels are [otherwise] low."

II. The Effect of Education on Fertility and Infant Health

The neoclassical model of fertility and child investment outlined above suggests that infant health may be thought of as a general function of inputs, some of which are choice variables of the mother and some of which are not. Formally, we model

$$Y = f(X, W)$$

where Y is a measure of the health of a particular mother's newborn child and X and W are vectors of inputs.[10] The vector X denotes inputs affecting infant health that are choice variables of the mother. For example, whether a mother smokes or drinks while pregnant, the extent to which she seeks prenatal care, and the quality of the man she has chosen as the father of her child—all of these are choice variables potentially affecting infant health. The vector W denotes endowments, which are immutable from the mother's perspective. These endowments include a mother's genetic ability to conceive and give birth to a healthy child and the permanent income and health knowledge of the newborn's grandparents.

While a woman's schooling choice cannot affect endowments, W, her schooling could affect her health inputs, elements of X. Hence, we model the demand function for health inputs as a function of schooling, endowments, and resources:

$$X = g(S, I, W)$$

where S denotes maternal schooling and I are resources. X and W are defined as above. Resources are meant to be interpreted broadly as non-schooling factors that affect a mother's choice of health inputs. One important such resource is income. For example, the quality and amount of prenatal care a woman receives is likely responsive to her income. Combining these equations, we have

$$Y = f(g(S, I, W), W) \tag{1}$$

In this model, the relationship between schooling and infant health is an admixture of the partial derivatives of the production function $f(\cdot)$ and the demand function $g(\cdot)$:

$$\frac{\partial Y}{\partial S} = \frac{\partial f}{\partial X}' \frac{\partial g}{\partial S} + \frac{\partial f}{\partial X}' \frac{\partial g}{\partial I'} \frac{\partial I}{\partial S} \tag{2}$$

Thus, the effect of education on infant health is inherently a reduced-form parameter summarizing (1) the impact of schooling on health inputs ($\frac{\partial g}{\partial S}$), (2) the impact of schooling on resources ($\frac{\partial I}{\partial S}$), (3) the impact of resources on health inputs ($\frac{\partial g}{\partial I'}$), and (4) the impact of health inputs on infant health ($\frac{\partial f}{\partial X}$).

[10] This parallels treatments by several authors. See, for example Rosenzweig and Schultz (1983).

The first term of equation (2) is the direct effect of education, the focus of Grossman's (1972) model of health capital. In particular, it is the effect of education on health inputs via health knowledge and the ability to process information. The second term of equation (2) is the indirect effect of education. It represents the effect of education via resources. A woman's education may affect her income and her choice of mate, which in turn could directly alter her choice of health inputs.

In this paper, we are unable to distinguish between the direct and indirect effects. Nevertheless, this distinction is important. It highlights the potential for heterogeneous education effects as there are several mechanisms by which education could potentially improve infant health. In the present case, the effect of education may differ considerably based on whether the mother is still enrolled in school, as discussed above.

Compared to the effect of education on infant health, the effect of education on fertility may be more closely linked to the effect of schooling on behavior. We will model

$$M = h(S, I, W)$$

where M is a fertility measure, such as whether a woman will become a mother (extensive margin) or completed fertility, conditional on motherhood (intensive margin), or age at birth (birth timing). The gradient of fertility in maternal schooling in this model is just the gradient of the production function $h(\cdot)$. Consequently, it might be said that the neoclassical prediction that education lowers fertility is stronger than the neoclassical prediction that education improves infant health.[11]

II.A. Identification with Random Samples

Bridging from theory to empirical implementation, consider a partially linear approximation to equation (1) for mothers:

$$Y = \theta S + \kappa(W) + \nu \tag{3}$$

where θ is interpreted as an average of the reduced-form parameter described in equation (2), the function $\kappa(\cdot)$ is a general mapping from endowments to infant health, and the residual

[11] We recognize that fertility, like infant health, may be related to maternal choices, X. For example, condom use and smoking are two such choices. Arguably, these types of effects sit more naturally in the type of model outlined in equations (1) and (2). However, we nonetheless think that a woman's fertility is more subject to her control than is the health of any children she might bear. Consequently, we view the models above as useful abstractions, focusing attention on the key aspects of the problem.

ν is meant to capture idiosyncratic factors unrelated to schooling and endowments.[12,13,14,15]

In a non-experimental setting with observational data, identification is difficult because variation in schooling may reflect endowments such as parental resources or time preferences. For example, individuals observed to obtain many years of schooling may be those who inherit good health from their parents. Alternatively, well-educated individuals may be patient people who may invest much in their own health and the health of their offspring (Fuchs 1980). A natural and traditional approach to this selection problem is to control for endowments, as for example in a regression version of equation (3) with proxies for the W, perhaps entered in a flexible way given the unknown shape of $\kappa(\cdot)$.

However, this approach has important limitations. The notion of endowments is abstract; theory gives only moderate guidance regarding what variables might be appropriate controls. Moreover, accurate measurement of these controls is difficult, leading omitted variable bias to persist even with proxies. Finally, it is well-documented that schooling itself is measured with moderate error; this leads to measurement error bias, even in the highly optimistic case that all endowments are known and measured with no error. Thus, it is possible for regression adjustment to be a cure which is worse than the disease.[16]

To circumvent these identification difficulties, we employ a regression discontinuity strategy. To understand how this approach may provide identification, let R denote day of birth, or the number of days subsequent to the school entry date that a person is born.[17] That is, a woman born on the school entry date is assigned $R = 0$, a woman born 10 days after the entry date is assigned $R = 10$, and a woman born 182 days before the entry date is assigned $R = -182$.

Figure 1 gives a schematic overview of this basic idea. Panel A shows a hypothetical conditional expectation of schooling[18] in day of birth for a cohort of children born within 182 days of the school entry date in a specific year (e.g., September 1, 1994).[19] Panel B also shows a hypothetical conditional expectation of schooling, but measured at the time of

[12] As θ is intended to capture the reduced-form effect of schooling, we exclude controls for resources I in equation (3).

[13] We do not require $\kappa(\cdot)$ to be smooth. Moreover, the additive separability in equation (3) is not critical. In particular, one could be interested in a version of equation (3) that included interaction terms between schooling and endowments. However, these interactions are often complicated to present and are moreover unusual in the literature. Thus, we view equation (3) as only a mild restriction and focus on it as the key object of interest for estimation.

[14] We discuss identification and estimation in the context of the effect of schooling on health. The effect of schooling on fertility proceeds analogously, replacing the outcome Y in equation (3) with a fertility outcome.

[15] For now, we ignore sample selection issues pertaining to which women elect to become mothers. This issue is taken up in Section II.C, below.

[16] A discussion of this problem is provided in, for example, Angrist and Krueger (1999).

[17] Note that, as defined, R has support $\{-182, -181, \ldots, -1, 0, 1, \ldots, 181, 182, 183\}$. (Recall that February 29th is a possible birthday.)

[18] Define schooling to be zero for kindergartners and one for first-graders.

[19] Such a cohort of children would have birthdates ranging from March 3, 1994 to March 2, 1995, and would begin first grade either in 2000 or 2001.

motherhood. This panel shows a discontinuity of much smaller magnitude than that in Panel A, because for many people, the precise timing of their birth relative to the school entry date will not affect how much schooling they have at motherhood (e.g., many women may have a target level of schooling and will successfully reach that target, regardless of their year of school entry). As such, we expect the discontinuity in education to be largest for women who are still enrolled in school. Panel C displays the hypothetical conditional expectation of low birth weight in mother's day of birth. Because schooling falls discontinuously in Panel B at the school entry date, and low birth weight is a negative measure of health, we expect to observe a higher incidence of low birth weight for women born just after the school entry date. The ratio of the discontinuities in Panels C and B estimates the effect of education on low birth weight. This estimate will have a causal interpretation, provided that education is the only thing changing discontinuously at the school entry date. To insure that this assumption is met, we can look at whether pre-determined characteristics exhibit a discontinuity at the school entry date. Panel D gives the hypothetical conditional probability of being African American in mother's day of birth. The smoothness of this function is interpreted as a corroboration of the assumptions underlying the regression discontinuity approach.

Our research design does not require that compliance with school entry policies be perfect. For instance, it would not violate our identifying assumptions if highly-educated parents encourage their children to start school early. Rather, the critical identification assumption is that within a narrow range of possible birthdays, one's exact birthday be randomly assigned. Perhaps the clearest way to think of the consequences of partial compliance for our study is by way of analogy with the randomized control trial, drawing on the local average treatment literature (Imbens and Angrist 1994).[20] If half of individuals are randomized at birth to receive a coupon paying for the first year of college, we would expect the treatment-control group education difference to be less than one, because some individuals will be unaffected by the treatment.

Moreover, even if those who are affected are unrepresentative of the treatment group, there is a specific sense in which a causal effect of interest is identified. Some individuals would never attend the first year of college, regardless of treatment assignment (never-takers), and some individuals would attend the first year of college, regardless of treatment assignment (always-takers).[21] For these people, treatment assignment does not affect educational attainment. However, some individuals will be persuaded to obtain additional education by virtue of their treatment assignment (compliers). As emphasized in the literature, when the effect of education is different for different people, an instrumental variables estimator does identify a causal effect of education, but this causal effect is specific to the subpopulation of

[20] The literature's first formal expression of these arguments is Hahn, Todd and van der Klaauw (2001).

[21] We may fruitfully insist that there do not exist persons so recalcitrant as who would attend the first year if they had received no coupon, but who would not do so upon receiving the coupon (defiers).

compliers.

We view our research design as providing precisely the type of randomization just described. For children born near the school entry date, being born just after the entry date may be thought of as a tax that makes it more difficult to obtain any particular level of schooling. In this framework, the children of parents who insist that their children start school a year earlier than mandated by school entry policies may be thought of as always-takers. Children who would be held back regardless of birthday are never-takers.[22] Children of parents who mildly comply with school entry policies are compliers.

The considerations above suggest that our estimates may be most relevant for children from low-education backgrounds. This conclusion is reinforced to the extent that children from low-education backgrounds are more likely to drop out of school. To see this clearly, consider two types of women. The first type of woman has a fixed idea regarding the level of education she desires to obtain before beginning a family. A woman of this type would not be included in the local average treatment effect identified by our analysis, because pregnancy would occur after she attained the schooling target and thus, her birthday would hold little predictive power as to her education. A second type of woman might seek to begin a family when the time felt right, and pay little regard to how much education she had managed to complete up until that point. If she is still in school when she elects to begin her family, a woman of this type will have one more year of education if she happened to be born before the school entry date than if she had happened to be born subsequent to the entry date. These considerations lead us to view our empirical analysis as primarily identifying the effect of maternal education on infant health for the subpopulation of women whose pregnancy interrupted their schooling. We note that these women are the target of many interventions to increase schooling.[23]

II.B. Estimation with Random Samples

Implementation of our research design hinges on estimation of a trio of conditional expectations. First, we seek to estimate $E[S|R = r]$ in order to ascertain the size of the discontinuity in schooling at motherhood resulting from school entry policies. Second, we seek to estimate $E[Y|R = r]$ in order to identify the parameter θ up to scale. Third, we seek to verify that the conditional distribution of W given $R = r$ is smooth in r. As discussed below, a natural way to do so is to test for the continuity of $E[W_j|R = r]$, where W_j is an

[22] As with our hypothetical example above, we rule out the existence of defier parents. To fix ideas, consider a defier parent in Texas. Such a parent would want their child to begin kindergarten at age 5 and 0 weeks, had their child been born in the last week of August—yet would want their child to begin kindergarten at age 5 years and 51 weeks, had their child been born in the first week of September.

[23] More generally, a women whose dropout decision is a function of her age rather than her schooling attainment will be in the relevant subpopulation. Thus, the subpopulation additionally includes those who leave school due to impatience or other factors idiosyncratic to schooling attainment.

element of (or a proxy for an element of) the vector of endowments W; a distributional extension is to test for continuity of $E[1(W_j \leq k)|R = r]$ for each k in the support of W_j.

How best, then, to estimate these conditional expectations? The econometrics literature has emphasized the use of local polynomial techniques to estimate conditional expectations in the regression discontinuity design (see, for example, Hahn et al. 2001 and Porter 2003). This is hardly surprising given their well-documented good behavior at boundaries (see, for example, Fan and Gijbels 1996). However, as a practical matter, for relatively well-behaved conditional expectations, estimates based on local polynomial techniques differ little from those based on global polynomial estimators (with a somewhat higher order of polynomial). This is certainly true in this application, since many of the conditional expectations we examine appear to be flat or linear in day of birth.

Consequently, we follow the recent applied regression discontinuity literature (e.g., Angrist and Lavy 1999, DiNardo and Lee 2004) and model the conditional expectations of infant health and schooling by parsimonious global polynomial methods.[24] Formally, we consider the simultaneous equations model

$$Y = \theta S + \alpha_0(P)R + \alpha_1(P)DR + \varepsilon \tag{4}$$

$$S = \pi D + \beta_0(P)R + \beta_1(P)DR + u \tag{5}$$

where $D \equiv 1(R \geq 0)$ and by way of analogy with the time series literature $\alpha_0(P)$, $\alpha_1(P)$, $\beta_0(P)$, and $\beta_1(P)$ denote low-order polynomials in P of the same order (interpret P^jR as R^j and P^jDR as D^jR^j for $j = 0, 1, \ldots$). For example, suppose that we take $\alpha_k(P)$ to be a third-order polynomial of the form $\alpha_{k,0}P^0 + \alpha_{k,1}P^1 + \alpha_{k,2}P^2 + \alpha_{k,3}P^3$ for $k = 0, 1$, and $\beta_k(P)$ analogously. Then the system above is a succinct way of writing

$$Y = \theta S + \alpha_{00} + \alpha_{01}R + \alpha_{02}R^2 + \alpha_{03}R^3 + \alpha_{10} + \alpha_{11}DR + \alpha_{12}DR^2 + \alpha_{13}DR^3 + \varepsilon \tag{4'}$$

$$S = \pi D + \beta_{00} + \beta_{01}R + \beta_{02}R^2 + \beta_{03}R^3 + \beta_{10} + \beta_{11}DR + \beta_{12}DR^2 + \beta_{13}DR^3 + u \tag{5'}$$

(Recall the convention $P^0DR = D^0R^0 = 1$.) The regression discontinuity design suffers from inherent ambiguity regarding the appropriate order of the polynomials in equations (4)-(5).[25] This is because the degree of smoothness in confounders required for identification is left unspecified—in short, the identifying assumptions are non-parametric in nature, and inherit the advantages and disadvantages of a non-parametric analysis. Throughout our empirical analysis, unless otherwise specified, we focus on interacted third-order polynomials in the interest of parsimony.[26]

[24] An earlier draft of this paper used local linear techniques exclusively. The results using these two approaches are quite comparable qualitatively. We opt for global polynomials here in order to focus the discussion on substantive issues. This has additional computational advantages in regards to ratio estimation and hypothesis testing.

[25] In the local polynomial context, there is an analogous ambiguity regarding the bandwidth parameter.

[26] As noted by Fan and Gijbels (1996) generally and Porter (2003) specifically in the regression discontinuity

In the system (4)-(5), the parameter θ measures the causal effect of schooling on infant health, and the parameter π measures the strength of the "first-stage" relationship between maternal education and age at school entry policies. Identification of θ requires continuity of the mapping from day of birth to endowments. In the simultaneous equations model of equations (4) and (5), identification of θ requires that D be orthogonal to ε, conditional on the polynomials in R and DR. This may be understood as assuming that for individuals born near in time, the precise date of their birth be random.

This assumption is partially testable, at least on the basis of observables. Specifically, all pre-determined characteristics of the mother should exhibit no discontinuity at the school entry date.[27] This is true in terms of average characteristics as well as distributions. We may formalize these ideas as follows. The assertion that D is a valid instrument for schooling is equivalent to the assertion that the sequence λ_k is zero in the sequence of regression models

$$1(W_j \leq k) = \lambda_k D + \xi_{0,k}(P)R + \xi_{1,k}(P)DR + \upsilon_k \tag{7}$$

where W_j denotes an element of or a proxy for an element of W, and the index k ranges over the support of W_j.[28] Thus, we are able to test formally whether mothers are systematically different along several education-independent dimensions by ascertaining whether the distributions of such characteristics are different for mothers born before and after the school entry date. In the empirical work to follow, we present estimates of equation (7) for a variety of proxies W_j. We also engage in the more traditional overidentification test of verifying the smoothness of the conditional expectation of pre-determined characteristics in birthday. Our results uniformly support the identifying assumptions. Stated plainly, maternal schooling (and those outcomes which could logically respond to schooling) are the only attributes of mothers which appear to vary discontinuously at the school entry date.

II.C. Identification with Selected Samples

As outlined above, estimation of the parameter θ in equation (4) is straightforward using random samples from the triple (Y, S, R). However, we focus on birth certificate data. These data are not random samples of the population, as we only observe women who give birth. Additionally, we only observe these mothers at their chosen age at birth, adding to the selection issues of the analysis.

Assessing the severity of selection on age at birth is straightforward because age at birth is recorded on birth certificates. That is, we may estimate the conditional expectation of

context, a cubic entails no more asymptotic variance than a quadratic and is likely to result in a smaller asymptotic bias. This result is general, leading to a mild preference for odd-order polynomials.

[27] Because assignment to starting grade occurs so early in life, pre-determined characteristics of mothers are essentially limited to those which are immutable (such as race) and those which are measured at birth.

[28] The polynomials $\xi_{0,k}(P)$ and $\xi_{1,k}(P)$ are constrained to be of the same order as $\beta_0(P)$ and $\beta_1(P)$, respectively. That is, the order of the polynomial model used in equation (5) must be sufficiently rich as to capture the fluctuations in the conditional probability that $W_j \leq k$.

maternal age in day of birth using the polynomial model in equation (5), after replacing the dependent variable of educational attainment with age at birth. If maternal age is continuous in day of birth, then selection is not a problem. However, if maternal age is discontinuous in day of birth, then the interpretation of any discontinuity in infant health at the school entry date is problematic—in principle, the infant health discontinuity could reflect differences in education or in age at first birth. The common practice in the literature appears to be to control for age. However, the model outlined in equations (1) to (2) suggests that age is a choice variable plausibly affected by the educational intervention.

This subtlety of interpretation is of little practical consequence here, as we find little evidence that age at birth is affected by school entry policies. That is, we show that the conditional expectation of maternal age given day of birth is smooth. More generally, we demonstrate that the distribution of age at birth is equal for mothers born just before and just after the entry date. Therefore, whether we control for age in the estimation of equations (4) and (5) does not affect our estimated infant health effects.

A more subtle identification problem we confront in using birth certificates is due to selection based on who decides to become a mother. To understand why this presents an identification problem, consider an augmented version of the reduced form of the system of equations (4) and (5), viewing Y as a latent infant health measure (conceptually defined as the health of a woman's child at birth, were she to become a mother), S as latent maternal schooling (conceptually defined as the amount of schooling a woman would have at motherhood, were she to become a mother) and M^* as a latent index determining motherhood:

$$S = \pi D + \beta_0(P)R + \beta_1(P)DR + u \tag{8}$$

$$Y = \rho D + \gamma_0(P)R + \gamma_1(P)DR + v \tag{9}$$

$$M^* = \tau D + \delta_0(P)R + \delta_1(P)DR + w \tag{10}$$

where we assume that motherhood occurs if $M^* \geq 0$, the parameter θ relates to those in the reduced form model by $\theta = \rho/\pi$, and the triples (Y, S, R) are only observed for mothers, or those with $M \equiv 1(M^* \geq 0) = 1$. Based on institutional details we expect $\pi \leq 0$, or less schooling at motherhood for those women born after the entry date. Based on economic theory we expect $\rho \leq 0$ (i.e., worse health for a woman's progeny due to lack of education) and $\tau \geq 0$ (i.e., greater likelihood of becoming a mother due to lack of education).

With a random sample of measures (Y, S, M, R), least squares is consistent for π, ρ and τ. However, we observe (Y, S, R) only for the selected sample of women choosing to become mothers. As shown by Heckman (1976) and emphasized in a long subsequent literature (see Moffitt (1999) and Das, Newey and Vella (2003) for discussion and references), with selected samples, identification will be difficult if the stochastic residuals u, v, and w are related.[29]

[29]Heckman pointed out that in selected samples, consistent estimates of the parameters in equation (8)

We are particularly interested in the possibility that $\tau = 0$ in the above system. In that special case, the latent index determining motherhood is equal for two women born just before and just after the school entry date. This leads immediately to a conclusion that π and ρ are consistently estimable—even in the selected sample of mothers—using least squares techniques. This follows because if $\tau = 0$, then the conditional probability of motherhood is a continuous function of day of birth (for proof, see Appendix I). Intuitively, if we are equally likely to observe women born just before and just after the entry date, then any sample selection effects difference out.[30]

If $\tau \neq 0$, then an alternative approach to identification is to augment equations (8) and (9) with polynomials in estimates of the probability of motherhood, perhaps from another data set. This is essentially the estimator proposed by Ahn and Powell (1993).[31] Because our estimates of the probability of motherhood are smooth near the school entry date, this more complex procedure would not materially alter our infant health estimates.

II.D. Estimation with Selected Samples

As mentioned above, we are particularly interested in estimation of τ, since the effect of education on infant health is identified in the selected sample of mothers if $\tau = 0$. At first glance, this would seem to be a useless result, since we cannot estimate τ using birth certificate data—these data do not contain any information on those women with $M = 0$, by definition. We are able to circumvent this problem due to the grouped nature of the model in equation (10). We next describe the auxiliary information that makes estimation possible and introduce notation to focus the discussion.

We augment our birth certificate data with information on the number of female children born daily from 1969 to 1988 (exact date of birth of the infant is available in the public-use Natality Detail Files from 1969 to 1988). Let $r \in \{$-182,-181,...,-1,0,1,...,182,183$\}$ index possible birthdays relative to the school entry date, or the support of R. For each r, let n_r denote the number of women born during this period with birthday r, and let m_r denote the number of women observed having a first birth in our birth certificate data from the same cohort and with the same relative birthday r. Then the relative frequencies $\widehat{p}_r = m_r/n_r$ estimate the probability of motherhood and may be used to estimate the parameters of equation (10).

could be obtained by regressing S on the variables on the right-hand side of those equations as well as a control function estimating the bias term $E\left[u|R, w > -\overline{M^*}\,\right]$, where $\overline{M^*} = \tau D + \delta_0(P)R + \delta_1(P)DR$. However, as the literature has emphasized, this approach works best when equation (10) is augmented by a variable which is uncorrelated with u and v. When no such variable is available, then the approach relies on functional form assumptions, which should be viewed with skepticism in this context, given the ambiguity surrounding parameterization of the polynomials.

[30]For the result to hold, the constant coefficients in equations (8) to (10) may be dispensed with, but the monotonicity implicit in equation (10) is critical (see Lee 2003).

[31]For an empirical application using this approach, see Card and Payne (2002).

A limitation of the simple ratio approach is censoring. To fix ideas, consider the case of right-censoring in Texas, where the administrative data we use spans 1989 to 2001. For a woman born in 1979, we will observe motherhood if her age at first birth occurs between ages 10 and 21.[32] If her age at first birth is 22, then the probability that we observe her birth declines monotonically in her birthday, with a woman born on January 1st being more likely to be observed than a woman born on December 31st. If she is older than 22 at birth, we do not observe her birth. Left-censoring is a problem for women in older cohorts, and both left- and right-censoring may be a problem for women in intermediate cohorts.

An easy solution to this problem is to include a woman in the count of the number of mothers (i.e., the numerator, calculated from administrative data) only if her delivery date is subsequent to her birthday in 1989, and prior to her birthday in 2001.[33] Compared to the simple ratio approach, this adjustment eliminates a spurious discontinuity at January 1st, which occurs because women born near the end of the year have a different probability of being observed compared to women born at the beginning of the year. This "rolling window" approach is most important for California, since January 1st is more proximate to December 1st, the school entry date for California, than it is to September 1st, the school entry date for Texas. We adopt this approach for both California and Texas, estimating the probability of motherhood using the adjusted procedure described, and then regressing this first-step estimate on D and a polynomial in R.

III. Data and Sample

In our analysis, we use confidential 1989-2001 Texas and 1989-2002 California natality data, acquired from each state's Department of Health.[34] We focus on recent natality data since the standard birth certificate started collecting the mother's exact date of birth beginning in 1989. Such information on the mother's exact date of birth is suppressed on the public-use national Natality Detail Files compiled by the National Center for Health Statistics and is usually considered identifying information. By special permission, we obtained access to a restricted version of the California and Texas data files.

[32] Provided she gives birth in her own birth state. We discuss (and reject) the possibility of endogenous mobility below.

[33] To be specific about our procedure, let i index all mothers in the Texas administrative data, R_i a mother's birthday relative to the school entry date, C_i a mother's delivery date, and $B_{i,1989}$ the date of a mother's birthday in 1989 and $B_{i,2001}$ the date of a mother's birthday in 2001 (measure all three dates in days since a shared date of reference). In this notation, the simple numerator count is $m_r = \sum_{i=1}^{n} 1(R_i = r)$, and the adjusted numerator count is $\widetilde{m}_r = \sum_{i=1}^{n} 1(R_i = r)1(B_{i,1989} \leq C_i < B_{i,2001})$. The probability of motherhood may then be estimated $\widetilde{p}_r = \widetilde{m}_r / n_r$. This adjustment reduces mechanically the level of the estimate of the probability of motherhood, but does so equally for all birthdays.

[34] There is no compelling reason why we include an extra year of California data. Our original analysis focused exclusively on Texas and we later added California mothers to our analysis. When we did so, the 2002 natality data for California had been released.

These natality files cover the universe of all births occurring in these states, approximately 800,000 births per year. At birth, each mother along with her health care provider completes an extensive survey, which inquires about maternal and paternal demographic characteristics, maternal behaviors during pregnancy (e.g., prenatal care), and the healthiness of the infant at birth.

For Texas, but not for California, our natality data are merged with infant mortality information from death certificates for those infants who died within the first year.[35] The background information contained in the death certificate is similar to that available in birth certificate. As a consequence, for Texas, we are able to examine whether maternal education affects infant mortality.

We impose three main sample restrictions. First, our sample consists exclusively of mothers born in the state in which they gave birth. An ideal analysis would use information on the state in which a mother began her education. We view state of birth as a reasonable proxy for the state where education begins.[36] This sample selection rule excludes the large number of mothers born abroad, for whom we do not know the school entry policies.

Second, we limit our sample to mothers who are 23 years old or younger. As noted above in Section II.A, the education discontinuity induced by school entry policies may mechanically close as schooling is completed. We observe only small differences in education at the entry date for women age 24 and above.

Third, we focus on first-time mothers. Without such a restriction, we would be worried that a mother having multiple births would appear in our sample more than once, leading our confidence intervals to be too small. We are unable to identify perfectly which mothers are repeated. Thus, to avoid this potential problem, we eliminate all second and higher older births from our sample.[37]

Our other sample restrictions affect the estimation sample only slightly. We exclude non-singleton births as the meaning or significance of infant health measures such as low birthweight vary by plurality.[38] For instance, the birthweight of most twins is below average, but this is not often viewed as an indicator that the infant has a health risk. Finally, we purge all mothers missing information on education and day of birth, the two main necessary elements of our research design.

[35] It is possible to acquire infant mortality data for California. However, the California Department of Health and Human Services only matched the natality and mortality data for selected years.

[36] Ideally, we would like to estimate the fraction of women born in California (Texas) giving birth in California (Texas) who began their education in California (Texas). We do not know of an auxiliary data set which measures state of onset of education and consequently, have no direct evidence on this question. However, it is relevant to note that in 2000, 89.5% of 5 year olds born in California still lived in California, and 89.8% of 5 year olds born in Texas still lived in Texas. These estimates almost surely understate the fraction of the mothers in our sample who began their education in their birth state.

[37] This has the added benefit of being a more homogenous sample as well as making our sample comparable with other results in the literature.

[38] Singleton births comprise approximately 98% of all births.

Before showing some descriptive statistics of these data, it is important to note that the quality of the data varies across states due to the different structure of the two birth certificates. This is most notable for risky maternal behaviors. In Texas, women are asked explicitly whether they smoked during pregnancy. In contrast, in California, maternal smoking must be inferred from a single question recording a variety of pregnancy conditions, including smoking. The question uses a series of numeric codes to describe these conditions. As an example, three such codes pertain to tobacco use, premature labor, and amniocentesis, which are arguably unrelated conceptually. Mothers self-report up to 16 such conditions (from among 30 possible codes). Smoking's measurement is thus very indirect. As we will see, the measured rate of California maternal smoking is only one-third that for Texas, which is suggestive of extensive underreporting. For these reasons, we believe that the reporting of smoking behaviors and the incidence of STDs is more reliable for Texas than for California.[39] On the other hand, to the extent that measurement errors are smooth in day of birth, our discontinuity estimates will be substantially unaffected.

Table 1 provides descriptive statistics for our study states. Throughout most of our analysis, we examine Texas and California separately. To get a sense of how selective is our main estimation sample (i.e., mothers who are born in the state in which they give birth, are having singleton first births, and are less than 24 years old), we present summary statistics for the overall sample of mothers with singleton births (first column) and the sample of first-time mothers with singleton births (second column) in addition to our young mothers sample (third column).

Relative to both all mothers and first-time mothers, our estimation sample is negatively selected. The first-time young native mothers are considerably younger, have lower levels of education, and worse birth outcomes. Compared to California mothers, Texas mothers have reduced number of years of schooling, are younger, and have a higher incidence of low birth weight birth but nearly equal likelihood of a premature birth. In terms of race and ethnicity, African American mothers comprise 14% of our main estimation sample for California and 20% of that for Texas, and for both states roughly 40% of the mothers in our main estimation sample are Hispanic.

IV. Results

We present our results in six subsections. First, we examine the impact of school entry policies on education at motherhood. These effects are stark and substantial. Second, we consider the impact of school entry policies on fertility. We find no regular pattern in day of birth for our two key fertility outcomes: the probability of becoming a mother and age at first birth. Third, we examine the impact of school entry policies on infant health, as proxied

[39] Maternal drinking is not one of the available pregnancy conditions, and we do not observe maternal drinking for California.

by birth weight, gestational length, and infant mortality. As with the fertility results, we find little evidence of any day of birth pattern. In the fourth subsection, we use the reduced form parameters of the preceding sections to calculate instrumental variables estimates of the effect of maternal education on infant health. In the fifth section, we examine the impact of school entry policies on several risk factors for poor infant health. Chief among these is the effect on the quality of a woman's mate, as proxied by father's age and education. Because father's education is a response to the manipulation of maternal education, we also present instrumental variables estimates of the effect of *total* parental education on infant health. Finally, in the sixth section, we conduct some robustness checks to insure that our research design manipulates only maternal education. These results confirm the expected smoothness in mother's day of birth for a variety of potential confounding variables.

IV.A. School Entry Policies and Education

We begin with a graphical presentation of the effect of school entry policies on education at motherhood. Figure 2 gives the relationship between schooling and day of birth separately for California and Texas. In both panels, the solid line in the figure is a parametric estimate of the conditional expectation of schooling at motherhood given date of birth. The parametric estimate corresponds to least squares fitted values corresponding to equation (5), where $\beta_0(P)$ and $\beta_1(P)$ are both taken to be third-order polynomials, and where additional control variables have been included for variance reduction only. The control variables are (i) indicators for race, (ii) indicators for ethnicity, (iii) indicators for age, and (iv) indicators for year of data. Below, we demonstrate that these controls are smooth functions of mother's day of birth.[40]

To help the reader gauge the sensitivity of our analysis to the parametric assumptions we place on the conditional expectation, we augment the parametric conditional expectation estimate by local averages, presented as open circles in the figure. These are regression-adjusted means of education for each day of birth.[41] It is important to note that (i) the deviation of the local averages from the parametric fit is a metric for the uncertainty regarding the true conditional expectation, and (ii) any reasonable estimator for the conditional expectation must pass through these local averages in the same way that the fitted values from a bivariate regression must pass through the scatterplot of the independent versus dependent variable.

Several aspects of the figure are worth noting. First, for young mothers in both California and Texas, there is a marked discontinuity in education at motherhood exactly at the school entry date, as expected. The size of the discontinuity is on the order of one-seventh of a

[40] Because of this smoothness, when we do not include these controls, our point estimate of the discontinuity is essentially unaffected.

[41] For the precise method of calculation and some technical details, see Appendix II.

year in California and one-fourth of a year in Texas. To interpret this magnitude, it is useful to assume that, for some fraction of persons, being born after the school entry date leads them to obtain one fewer years of schooling than they otherwise would have, and that for all other persons, schooling is unaffected by birth day relative to the school entry date. Under this assumption, the fraction of young mothers who are affected by school entry policies may be estimated by a normalized education discontinuity at motherhood, where the normalization is by the initial compliance rate with school entry policies (cf. Figure 1, Panel A). Matsudaira and McCrary (2004) estimate that the initial discontinuity in education is about four-fifths. Assuming this estimate is valid, the fraction of young mothers whose education at motherhood is affected by school entry policies is a large one-fifth for California and one-third for Texas.[42]

A second noteworthy aspect of the figure is that there any no evidence of a discontinuity at dates other than the school entry dates. The juxtaposition of the smoothness of the conditional expectation away from the school entry date and the sharpness of the discontinuity at the entry date supports the interpretation of the education discontinuity as directly attributable to school entry policies. Third and finally, this inference is particularly compelling since the school entry date is different for the two states—December 1st for California and September 1st for Texas.

For reference, point estimates for the discontinuities in the figure are given in Table 2. Standard errors are presented in parentheses and are clustered standard errors at the level of day of birth.[43] The t-ratio on the estimated discontinuity for California is over eight, while that for Texas is over twenty, suggesting that this instrument works well enough that we ignore statistical issues with finite sample bias (Bound et al. 1995, Staiger and Stock 1997) and size distortions of Wald test statistics (Moreira 2002).[44]

An interesting pattern in the data that is obscured by the averages in Figure 2 is that the education discontinuity is strongest for the youngest mothers and weakest for the oldest mothers. Figure 3 provides separate education discontinuity estimates for each age at birth from 15 to 35. That is, we estimate equation (5) separately for the sample of 15-year-old mothers, 16-year-old mothers, and so forth. We supplement these disaggregated regression discontinuity estimates with female school enrollment estimates for our two study states, calculated from the 5% 2000 public-use microsample (Census Bureau 2003). The figure shows discontinuity estimates that decline in magnitude with age as enrollment rates fall.

This simple pattern is consistent with two stories. One story is that, for women in

[42]For a similar analysis, see Angrist and Krueger (1992).

[43]For both states, the point estimate and standard error are similar to those that result from running a group-level regression of the 366 local averages on the polynomial model specified. Additionally, the clustered standard errors are similar to standard errors that assume a spherical variance matrix. Further evidence on robustness is given in Appendix II.

[44]The t-ratios here are very similar in magnitude to those on the exclusion of the instrument in the first-stage regressions of Currie and Moretti (2003).

the cohorts we study, school entry policies have no impact on *completed* education, but do manipulate education at motherhood for those whose pregnancy interrupts their schooling. A second story is that, for women in these cohorts, young mothers are precisely those who drop out of school as soon as possible, and that older mothers are exactly those whose educational attainment would not be affected by when they started their schooling because they stop schooling based on completed schooling rather than their age. If the first story is more apt, then the best interpretation of the fertility and infant health effects we estimate are the effects of education, holding income constant, or the effect of education via learning. This follows because a woman who anticipates returning to school may not perceive her permanent income to be reduced by virtue of having little education at the time of her birth. If the second story is more apt, then the best interpretation of the effects we estimate are the effects of education, including the effect of education on permanent income (assuming foresight). Compared to the first case, in the second case we would expect a stronger influence of education on fertility and infant health.

An important issue for interpretation of our estimates is the education level for which school entry policies have the biggest effect. For example, the approach of Angrist and Krueger (1991) manipulates primarily the number of years of high school an individual completed, whereas the approach of Currie and Moretti (2003) affects the probability of attending college. In the case of a binary instrument for schooling, $Z \in \{0, 1\}$, estimation of the affected educational attainments is easily addressed by examining the vertical differences in the empirical distribution function of schooling for observations with $Z = 1$ and $Z = 0$ (e.g., pre- and post-cutoff mothers, respectively).[45]

An artificial way to recover this function is to estimate, for each level of schooling s, the parameter η_s in the regression

$$1(S \leq s) = \phi_{0,s} + \eta_s Z + \nu_s \tag{11}$$

This suggests a natural extension to the regression discontinuity context: estimate a sequence of regressions of the form

$$1(S \leq s) = \eta_s D + \phi_{0,s}(P)R + \phi_{1,s}(P)DR + \nu_s \tag{12}$$

paralleling the modeling of equation (8).

Figure 4 plots as solid circles the pairs $(s, \widehat{\eta}_s)$, where $\widehat{\eta}_s$ is the least squares estimate of η_s in equation (12). In each regression, we include the same set of control variables we used in estimating the conditional expectations of education plotted in Figure 2. As we are examining a set of point estimates rather than a single point estimate, we present uniform confidence regions rather than pointwise confidence regions (for details, see Appendix II).

[45] Angrist and Imbens (1995) refer to this difference as the average causal response weighting function, because it gives the implicit weights underlying the probability limit of the instrumental variables estimator.

This is analogous to testing the null hypothesis that all coefficients are jointly zero in a least squares regression, where a joint F-test is preferred to a sequence of t-tests.

Examining the estimates in Figure 4, it is clear that the bulk of the variation in education induced by school entry policies is in the high school years. However, it would be incorrect to say that there is no effect beyond high school. For both California and Texas, there is an apparent effect on thirteen years of education, and even on fourteen years, which corresponds roughly to an associate's degree. For Texas, the effects appear to persist even into the third and fourth years of college, with effects at sixteen years on the margin of significance. A natural interpretation of this pattern is that, particularly for women in Texas, pregnancies interrupting educational progression occur with some frequency during college as well as during high school. The educational levels manipulated are strikingly similar to those documented in Angrist and Imbens (1995).

IV.B. School Entry Policies and Fertility

The effect of education on fertility could manifest itself in terms of the probability of ever becoming a mother, the number of children, and the timing of childbearing. As discussed below, for several cohorts of women we observe a direct estimate of the probability of becoming a mother. We do not observe completed fertility, as our sample frame is too short. However, we observe age at birth, a fertility timing measure.

Our measure of the probability of motherhood is the fraction of women born daily in California and Texas from 1969 to 1988 who are observed giving birth in California and Texas, respectively, between 1989 and the present. We have adjusted these probabilities in a minor way to circumvent censoring (see Section II.D). These estimates are presented in Figure 5. Each open circle in this figure is an estimated probability of motherhood for a specific birthday.[46] The solid line in the figure is a parametric estimate based on seventh-order polynomials fitted separately to either side of the entry date.[47] In addition, the figure includes a smaller, inset figure giving the probability of motherhood for each possible

[46] In the notation from Section II.D, these correspond to $\widetilde{p}_r$.

[47] We use interacted seventh-order polynomials because the fit of the interacted third-order polynomials near the entry date was poor. We also estimated lower and higher orders of polynomials; these were similarly negative and sometimes significantly different from zero. However, this is not strong evidence of a true discontinuity: point and standard error estimation for discontinuities is difficult in the presence of strong derivatives, as a general proposition, and the figure suggests fairly substantial seasonality in the probability of motherhood. Table A2 presents an F-statistic of the null hypothesis that the fit of the local averages is no better than the fit of the polynomial model. This F-statistic rejects the null hypothesis, and we obtain similar rejections for both California and Texas for (interacted) odd- and even-order polynomials of orders three through nine, inclusive. See Lee and Card (2004) for a standard error adjustment that is appropriate in this instance under an auxiliary assumption of normal misspecification errors. We elected to use a seventh order polynomial on a subjective basis, choosing the polynomial that best managed to reflect the local averages near the school entry date.

birthdate from January 1, 1969 to December 31, 1988.[48] The inset figure demonstrates clearly the effects of censoring. For cohorts born in the early 1970s, we observe their fertility behavior in their twenties and early thirties. This means that, for women in these cohorts, we are very likely to observe their first birth: at the peak, we observe births for 54% of California women and 62% of Texas women. However, for cohorts born in the late 1980s, we observe only a very small fraction at their first birth.[49]

Several aspects of the figure are notable. First, there is no decisive visual evidence of any discontinuity in the motherhood proxy near the cutoff date. Second, compared to the conditional expectation of schooling, the conditional expectation of motherhood exhibits substantial seasonality.

The point estimates for the effect of school entry policies on the probability of motherhood for both California and Texas are small in economic terms and precisely estimated (Table 2).[50] Recalling the neoclassical prediction that education reduces fertility, it is surprising that both point estimates are negative. However, neither effect is statistically distinct from zero.

These estimates of the probability of motherhood are biased to the extent that education affects mobility. In particular, if education increases the probability of emigration out of our two study states, we will be more likely to observe giving birth those women whose birthdays follow the school entry date. However, this would bias the motherhood estimates upward, making it more likely that we would find a positive effect. Therefore, we conclude that endogenous mobility, albeit in principle a valid concern, is not a dominant pattern in these data.

We next present estimates corresponding to the effect of school entry policies on age at birth. Figure 6 gives estimates of the conditional expectation of age at motherhood in own day of birth. These estimates are based on the polynomial model in equation (8) controlling for race, ethnicity, and year of birth, and replacing the dependent variable of schooling with age. Because the outcome variable is age, the sample includes all mothers (i.e., we do not restrict the sample to those aged 23 or younger). The results are substantively similar to those regarding motherhood in two regards. First, there is little visual evidence of a

[48] The inset figure uses the uncorrected estimates of the probability of motherhood, or $\widehat{p}_r$. This is because our censoring correction method *introduces* distortion into the conditional expectation given date of birth in order to eliminate distortion in the conditional expectation given day of birth.

[49] Using the Census, we can verify that these sampling fractions are reasonable. In particular, using the Census, we estimate the probability that a Texas-born woman aged 25 remained living in Texas on Census day 2000 to be 78.8%. Using the 1990 Census, Royer (2004) estimates that 86.9% of Texas women aged 45 and older had had at least one birth. If these estimates are reasonable approximations for the younger cohorts we study, then the product of these two numbers (68.5%) should be about the height of our probability of motherhood at its peak (62.5%). The minor discrepancy suggests either that the probability of motherhood is lower for more recent cohorts, that mobility is higher for more recent cohorts, or that there is still censoring even for the "peak" cohorts.

[50] Note that because the point estimate is derived from the group-level regression, no clustering is necessary.

discontinuity in age at the entry date. Second, Texas continues to exhibit much stronger seasonal fertility patterns than does California.

For both California and Texas, point estimates of the effect of school entry policies on age at birth (Table 2) are small in economic terms. For example, the estimate for California is consistent with a scant two percent of women born after the entry date giving birth one year later than women born before the entry date. Moreover, in statistical terms, neither estimate is distinct from zero, despite a sample size of nearly 1,300,000 mothers for California and nearly 900,000 mothers for Texas.

Although school entry policies do not affect average age at motherhood, they could nonetheless affect the likelihood of giving birth at different ages. A natural method for assessing this is to estimate the discontinuity in the probability of giving birth at age a or below, for a range of values a (i.e., to essentially estimate the distribution function of age at birth for those born just before and just after the school entry date). We implement this by using the same procedure used in analyzing mother's education.

We present these estimates in Figure 7, which parallels the analysis of Figure 4. The estimates for California are essentially zero for all ages. The estimates for Texas, while within sampling variation of zero for all but age fifteen, are somewhat low for the teenage years. We have three remarks in regard to this pattern for Texas. First, we note that the pattern indicates that children born after the school entry date have a *reduced* incidence of teenage pregnancy, despite having less education. That is, the sign of this effect is perverse. Second, we caution against interpretation of such subtle, statistically insignificant patterns, particularly given the seasonality of fertility in Texas. Third, we find no age effects if we replace the cubic polynomials in day of birth with quintic polynomials (results available from authors upon request). Looking at the evidence for California and Texas together, we feel the best conclusion is that there is little evidence of any impact of school entry policies on age at birth.

In summary, we find little evidence that school entry policies affect either the probability of motherhood or age at first birth. The implication of these conclusions is both substantive and statistical. Substantively, the lack of impact of school entry policies on these fertility outcomes can be interpreted as indicative of a limited causal role for education. Statistically, the lack of an impact of school entry policies on fertility means that birth certificate data—while in principle selected on both motherhood and age at birth—are essentially as good as a random sample, for the purposes of our analysis. This implies that our estimates of the education and infant health discontinuity are consistent, allowing us to estimate the effect of education on infant health without resorting to additional econometric complications.

Before turning to the estimated effect of these policies on infant health, we note that our fertility results are consistent with a simple biological model in which age of menarche is idiosyncratic, women begin having sexual intercourse soon after menarche and regardless of

grade, and some fraction of these women become pregnant (for a discussion of related ideas, see Akerlof, Yellen and Katz 1996).

IV.C. School Entry Policies and Infant Health

We begin our analysis of infant health with an examination of the impact of school entry policies on the incidence of low birth weight, one of the most widely cited risk factors for poor infant health.[51] Following our previous analyses, we begin with a graphical presentation of the day of birth pattern in the incidence of low birth weight. Figure 8 gives an estimate of the conditional expectation of low birth weight in day of birth, augmented by local averages, for our main estimation sample of young mothers.

If additional schooling reduces the incidence of low birth weight, as neoclassically predicted, we would expect to see an increased likelihood of low birth weight at the school entry date. However, the data do not support this prediction. In fact, the local averages suggest that low birth weight is largely idiosyncratic in mother's day of birth. Moreover, the gap in the conditional expectation at the school entry date is perversely signed: those born after the school entry date appear to have very slightly depressed rates of low birth weight, despite their lesser schooling.

Table 2 reports point estimates of the discontinuity in low birth weight. We estimate that mothers born after the entry date have a 0.14 percentage point reduced probability of a low birth weight birth, as compared to mothers born before the date. For Texas, we estimate that there is a reduced likelihood of 0.39 percentage points. For context, it is useful to recall from Table 1 that the overall rate of low birth weight is just over 6% for young California mothers and almost 8% for young Texas mothers. In percentage terms, the reduction in low birth weight attributable to birth after the entry date is not insubstantial in magnitude—a reduction of 2% in the rate for California and nearly 5% for Texas. However, both estimated discontinuities are statistically indistinct from zero. Thus, there is no evidence that the variation in education induced by school entry policies translates into differences in the incidence of low birth weight.

In addition to low birth weight, researchers are interested in other birth weight thresholds, at both tails of the birth weight distribution. For instance, researchers have been particularly interested in the incidence of very low birth weight (birth weight less than 1500 grams), very very low birth weight (birth weight less than 1000 grams), and high birth weight (more than 4000 grams). Given such an array of birth weight cutoff points, we examine the effect of school entry policies on birth weight distributions, just as we did for maternal education and maternal age (Figure 9). Since the region bracketed by the upper and lower limits of the uniform confidence region encompasses zero, we conclude that school entry policies exert no detectable impact on the probability of giving birth to a child of any particular weight.

[51] See Almond, Chay and Lee (2004) for references.

We next consider the impact of school entry policies on the incidence of premature birth, defined as a gestation length of less than 37 weeks. Figure 10 gives an estimate of the conditional expectation of prematurity in day of birth, augmented by local averages, for our main estimation sample. In the presence of an effect of education on prematurity, we would expect to see a rise in prematurity at the school entry date. However, as with low birth weight, the data do not support this prediction. The local averages indicate no particular pattern in day of birth, and the conditional expectation estimate again points to small perverse effects (Table 2).

For both California and Texas, we estimate that mothers born after the entry date have a reduced rate of prematurity, with a magnitude of about one-fifth of a percentage point. As the overall incidence of prematurity for these samples of women is roughly 10%, the implied percent reduction is below 2% for both states. These are more on the modest side than those for low birth weight, and are similarly within sampling variation of zero: the t-ratio for both estimates is roughly 0.6, and pooling the estimates raises the t-ratio to only 0.8.

As the choice of looking at 37 weeks as a cutoff point may seem arbitrary, we estimated the distribution of effects of school entry policies on gestation in weeks (results available from authors upon request). As with the analysis of birth weight, the statistically insignificant results hold throughout the gestation length distribution. Point estimates for various gestation lengths are sometimes positive and sometimes negative but well within the uniform confidence region.

Perhaps the most extreme measure of poor infant health is infant mortality. Figure 11 provides the pattern of infant mortality by mother's day of birth. Unlike the estimated effects for low birth weight and prematurity, the infant mortality estimate is not perverse in sign. We estimate that the discontinuity in the conditional probability of infant mortality is 0.06 percentage points (Table 2). Although this estimate may be viewed as numerically small, it is somewhat large relative to the baseline 6.7 infant deaths per thousand births for young Texas mothers (Table 1). However, the estimated effect is within sampling variation of zero, arguing against strong interpretation.

IV.D. The Effect of Education on Infant Health

So far, we have examined the reduced-form impact of school entry policies on education, fertility, and infant health. The evidence points firmly toward the conclusion that starting school later in life leads to

1. substantially reduced education at motherhood, particularly for young mothers,
2. negligible effects on the likelihood of giving birth by any particular age, and
3. negligible effects on infant health of this reduced schooling

However, a reduced-form analysis does not tell us what kinds of magnitudes regarding the effect of education may be ruled out by our analysis.

To address this, we turn to instrumental variables estimates of the effect of an additional year of maternal education on infant health (Table 3). We consider two types of instrumental variables estimates: those based on the regression discontinuity design (IV/RD) and those based on additonal exclusion restrictions (IV/DOB) which we discuss below.

We first discuss the IV/RD estimates. The table gives estimates of the effect of maternal education on low birth weight, prematurity, and infant mortality. These estimates are simply the ratio of the reduced-form estimates to the first-stage estimate of the effect of school entry policies on maternal education.[52] Standard errors, following the practice of Table 2, are calculated using clustered standard errors at the level of day of birth. Because each of the outcomes is a measure of ill health, the expected sign for all coefficients is negative (additional education should reduce the incidence of bad outcomes). Looking over the estimates in the table, however, we see that four of the five are positive.

A key reason for interest in these estimates is that they are likely to be consistent because the identifying assumptions are so plausible. However, as the estimates are based on narrow contrasts, they have a high degree of sampling variation. Nonetheless, several of the individual coefficients are estimated with sufficient precision to rule out magnitudes of economic interest. For example, the low birth weight estimate for California is 0.96 percentage points (standard error of 1.5 percentage points) while that for Texas is 1.5 percentage points (standard error of 1 percentage point). For Texas, using a one-tailed test of 5% size, we may rule out even the very small effect size of a 0.07 percentage point reduction.[53] For comparison, the recent results of Currie and Moretti (2003) suggest that a one-year increase in education reduces the rate of low birth weight by a full 1 percentage point. Our estimate for California is less precise, and rules out a 1.5 percentage point reduction in the rate of low birth weight.

As with the estimates for low birth weight, the prematurity estimates for both states are perverse in sign. For California, we estimate that an additional year of maternal education increases the incidence of prematurity by 1.3 percentage points (standard error of 2.1 percentage points). For Texas, the estimated effect is an increase of 0.7 percentage points (standard error of 1.2 percentage points). The IV/RD estimate of the effect of education on infant mortality is -0.25 percentage points. The baseline infant mortality rate for the mothers underlying this estimate is 6.7 per thousand births, so the point estimate is large in magnitude. However, the point estimate is also not precisely estimated—if the lower edge of the confidence region were the true parameter, then infant mortality could be eliminated by a one-year increment to schooling.

[52] That is, these are instrumental variables estimates using D as an excluded instrument for schooling, and a cubic in R and a cubic in DR as included instruments (see equations (4) and (5)).

[53] Throughout our discussion of magnitudes, we will use one-tailed tests of 5% size. This type of test is appropriate, since we seek to determine the negative value of largest magnitude that may be ruled out.

We next consider three different approaches to increasing the precision of these estimates. First, we pool the estimates for California and Texas (Table 4). The pooled IV/RD estimate for low birth weight remains of perverse sign (1.2 percentage points) and is estimated with some precision (standard error of just below 1 percentage point). This estimate rules out an effect size of -0.3 percentage points. The IV/RD estimate for prematurity is also of perverse sign and estimated with precision. For this outcome, we just fail to rule out a reduction in the incidence of prematurity of 1 percentage point (the Currie and Moretti point estimate for prematurity).

Our second approach to efficiency is to consider instrumental variable estimators based on additional exclusion restrictions motivated by Angrist and Krueger (1991) (labelled IV/DOB in Table 3). Angrist and Krueger assumed that individuals born in different quarters of the year are exchangeable. Bound et al. (1995) argue that there is substantial variation in births across seasons, and that this seasonality invalidates this exchangeability assumption.[54] A specific reason for interest in the identification strategy used in this paper is that it produces estimates which are robust to this criticism, provided that seasonality of birth is smooth (a testable assumption which we later assess).

We view the Angrist and Krueger assumption as controversial, but possibly correct. If we assume that the Angrist and Krueger assumption is correct, a more precise instrumental variables estimate of the effect of maternal education on infant health may be obtained. In the framework we develop in equations (4) to (5), the Angrist and Krueger assumptions correspond to assuming that the coefficients in the polynomials $\alpha_0(P)$ and $\alpha_1(P)$ in equation (4) are all zero. This assumption implies the consistency of (over-identified) instrumental variables estimates which use D, a polynomial in R, and a polynomial in DR as excluded instruments for schooling.

A quick comparison of the standard errors for the IV/RD estimates and the IV/DOB estimates (Table 3) reveals that the more precise IV/DOB estimates have confidence intervals that are approximately one-third (California) and one-half (Texas) the width of those for the regression discontinuity estimates. The IV/DOB estimates are generally closer to zero in magnitude than the IV/RD estimates and remain perversely signed. For example, the low birth weight estimate for both California and Texas is positive. The confidence region for California allows us to reject an effect size of -0.7 percentage points. For Texas, the estimate is positive and statistically distinct from zero (two-sided test), and taken at face value rules out the hypothesis that maternal education is protective of health.[55] For prematurity, the lower bound of the one-sided confidence interval for California is a reduction of one percentage point, and for Texas the lower bound is -0.75 percentage points. For infant mortality, the

[54] This point was argued in more detail in Bound and Jaeger (2000).

[55] In light of the large number of point estimates considered, we suspect that the best interpretation of the marginal statistical significance of the Texas estimate is chance.

IV/DOB estimate just excludes a reduction of one-third of a percentage point per year of maternal education. We note, however, that this may be a large magnitude in economic terms, even if not in statistical terms.

Because the IV/DOB estimates are over-identified, we may test the overidentifying restrictions formally using a J test. These are presented below the estimates, with associated p-values are given in brackets. The smallest of the five p-values is a reassuring 0.43, providing little evidence against the overidentifying restrictions imposed by the more precise IV/DOB estimates. On the other hand, these tests are known to have poor power properties.

A third approach to efficiency is to combine the California and Texas IV/DOB estimates (Table 4). These are the most precise estimates we consider. For low birth weight, we can use this approach to rule out even extremely small effect sizes—the lower bound of the one-sided confidence interval is -0.00028, or negative three-hundredths of a percentage point. For prematurity, the results are similarly precise, and we can reject effect sizes as small as negative seven-tenths of a percentage point.

Generally, the IV/DOB estimates appear to be of the same magnitude as the regression discontinuity estimates. This informal assessment may be made precise by means of a Hausman (1978) test (Table 3).[56] A quick glance at the table confirms that there are no important statistical differences between the two types of estimates, despite their different identifying assumptions.

Taken as a whole, we view the results in this section as highly suggestive of a limited role for maternal education in infant health, at least for the subpopulation of women whose education is affected by school entry policies. Our estimates generally rule out the point estimates from the existing literature, and the more precise estimates we consider rule out even very small effect sizes.

IV.E. School Entry Policies and Risk Factors

As emphasized in the theoretical discussion in Section II, the effect of maternal schooling on infant health is an admixture of a mother's behavioral response to schooling and the effect of her behavior on the health of her child. For this reason, it is important to attempt to flesh out which causal pathways from education to infant health are, or are not, implicated by our analysis. This may be particularly important in comparing our results to those in the literature, as it is not necessarily clear what the effect of education is supposed to represent.

To learn about these causal pathways, we estimate both reduced-form estimates of the impact of school entry policies on risk factors for poor infant health (Table 2) and instrumental variables (IV/RD and IV/DOB) estimates of the impact of maternal education on these risk factors (Table 5). The risk factors we consider may be thought of as falling into three key categories. The first category, which we term "risky maternal behaviors", encompasses

[56] See Davidson and MacKinnon (1993, pp. 241-242).

maternal smoking, maternal drinking, and maternal sexually transmitted diseases. The second category is comprised of several prenatal care measures. These are whether the mother obtained prenatal care at any point during her pregnancy, whether she obtained prenatal care during the first trimester, and the number of prenatal care visits she had. The third category of risk factors pertain to the quality of the infant's father. We view the father's quality as being higher (i) the more involved he is in the birth,[57] (ii) the older he is, and (iii) the more educated he is, because each of these factors contributes directly to household permanent income.

Looking at the reduced-form estimates in Table 2, we see that the estimates for maternal behaviors are of small magnitude, of mixed sign, and generally not statistically significant. These patterns are particularly interesting, in light of the impact of college openings on maternal smoking documented by Currie and Moretti. Turning to the estimates for prenatal care, we see that five of the six estimates are of the expected sign—mothers with less education are less likely to receive prenatal care, less likely to receive it early, and receive less of it. However, the estimates are of extremely modest magnitudes (cf. Table 1) and only the Texas estimate for care in the first trimester reaches the margin of statistical significance.[58]

In contrast, the paternal quality effects appear to be sizeable. These estimates show that women born just subsequent to the school entry date have mates who are younger and less educated, on average, than the mates of women born just before the entry date. The point estimates for both California and Texas are large and statistically distinct from zero. The statistical robustness of this conclusion is emphasized in Figure 12, which presents an estimate of paternal education in mother's day of birth.

Table 4 presents instrumental variables estimates (IV/RD and IV/DOB) of the effect of schooling on risk factors. For precision and to economize on space, we present pooled estimates only; these are calculated in the same manner as the instrumental variable estimates of the impact of education on infant health.

The estimates in Table 4 provide little evidence of any impact of education on maternal smoking or maternal STDs; the maternal drinking estimate (unreported) is similarly small in magnitude and statistically insignificant. However, the prenatal care results are somewhat suggestive. For example, both the IV/RD and IV/DOB estimates suggest that one year of education increases the probability of receiving care in the first trimester by 3 percentage points. The IV/RD estimate is not significantly different from zero, but the IV/DOB estimate has a t-ratio of over 5. Similarly, both the IV/RD and IV/DOB estimates suggest that

[57] We proxy involvement by the presence of the father's date of birth and educational attainment on the birth certificate rather than by marital status. Accurate measurement of marital status is a serious problem in the natality files. See Ventura and Bachrach (2000) for discussion.

[58] Below, we assess statistically how well the global polynomial fits the underlying data. We find that for some of these prenatal care outcomes, the global polynomial is inadequate, leading us to view the prenatal care results with some skepticism. For further details, see Appendix II.

one year of education increases the number of prenatal care visits by one-fifth. The IV/RD estimate is imprecise, but the IV/DOB estimate has a t-ratio of almost 4.

For paternal quality, both the IV/RD and IV/DOB estimates are fairly precise. A one-year increase in maternal education is predicted to cause an increase in paternal age of four-fifths of a year, and an increase in paternal education of one-third to over one-half of a school year. The mating-market spillovers suggested by these estimates are important for distinguishing partial and general equilibrium effects of a manipulation of female schooling (Behrman and Rosenzweig 2002, p. 334).

Both Behrman and Rosenzweig (2002) and Breierova and Duflo (2004) present evidence that paternal education may be as important as maternal education in regards to its impact on child outcomes. If this is correct, then the scaling of maternal and paternal education implicit in the estimates in Table 3 is somewhat misleading, and the more apt parameter of interest is the effect of *total* parental education on infant health. Instrumental variables estimates of this parameter, based on a system like that of equations (4) and (5) replacing maternal schooling with the sum of maternal and paternal education, are presented in Table 6.[59]

Looking at the IV/RD estimates of the impact of total parental education on infant health in Table 6, we see that all five of the estimates are perversely signed.[60] However, the magnitudes of these effects rarely exceed zero by more than sampling variation: the only statistically significant effect is that for low birth weight in Texas.

For the more precise IV/DOB estimates, three of the five estimates are of the "expected" sign, but (one-sided) confidence regions based on these estimates are quite tight. For example, for California, the lower edge of the one-sided confidence interval for low birth weight is -0.6 percentage points, and the lower bound of the interval for prematurity is -1 percentage point. For Texas, we would reject a low birth weight effect of -0.2 percentage points, a prematurity effect of -0.6 percentage points, and an infant mortality effect of -0.003 percentage points. For completeness, Table 6 also presents IV/RD and IV/DOB estimates of total parental education on the proxies for risky maternal behaviors and take-up of prenatal care that were presented in Table 5. The results for maternal smoking, drinking, and sexually transmitted diseases are small in magnitude and statistically insignificant. However, several

[59]The day of birth under consideration, however, continues to be that of the mother. We have also estimated extended versions of this model using father's day of birth in addition to mother's day of birth. These estimates are generally corroborative of those given here, but are imprecise. This is because the first-stage relationship between father's day of birth and educational attainment of fathers and mothers is much more muted than the analogous relationship for mothers. Indeed, only for Texas is the first-stage relationship significant at conventional levels (t-ratio on discontinuity of father's education of 3).

[60]The careful reader may be surprised that the infant mortality result is of a different sign than the results in Tables 2 and 3. This is the result of a slightly different estimation sample (by necessity, we restrict the sample to those observations with valid information on fathers). The IV/RD estimate for *maternal* education using the same estimation sample as underlies Table 6 is 0.0001 (standard error of 0.0041) and the IV/DOB estimate is -0.0009 (standard error of 0.0015).

of the prenatal care estimates are of large magnitude and estimated with some precision.

Given the range of risk factors affected by the educational manipulation of school entry policies, we are particularly surprised by the small magnitude of our estimated infant health effects. However, an important issue that remains unresolved is the validity of the research design, which is a precondition for interpretation. In the next section, we take up the issue of tests of overidentification. These tests uniformly support the identifying assumption that, local to the school entry date, the precise timing of a mother's date of birth is effectively random, giving credence to the education estimates just discussed.

IV.F. Testing the Validity of the Research Design

In this section, we present evidence that pre-determined maternal characteristics are smooth in day of birth. A practical difficulty with testing the smoothness of endowments is that assignment to starting grade occurs at birth, so that few attributes of the mother are pre-determined. Race and ethnicity are immutable, so they could not logically react to assignment to starting grade. Therefore, we may examine whether fraction black and fraction Hispanic are smooth in mother's birthday using our administrative data. However, all other maternal characteristics we observe may logically be viewed as a response to assignment to starting grade and therefore are not useful for testing the validity of the research design.

To put our assumptions to a strong test, we turn again to the public-use Natality Detail Files from 1969-1988. For mothers born in these years, we can verify the smoothness of a variety of maternal and grandparental characteristics. To economize on space, we present no figures and focus instead on point estimation of discontinuities.

The first row of Table 7 gives estimated discontinuities for selected pre-determined characteristics of mothers. Each entry is a discontinuity estimate for a different pre-determined characteristic, calculated in the same manner as for those in Table 2, but using no auxiliary controls. Standard errors, in parentheses, are clustered at the level of birthday. If the regression discontinuity approach is appropriate, then then there should little evidence of any discontinuity in pre-determined characteristics. F-statistics for this hypothesis are presented below the standard errors, and p-values are given in brackets.[61]

Scanning across the columns, there is little evidence of any discontinuity in the maternal characteristics we measure: fraction Hispanic, fraction black, and low and high birthweight.[62] This is true for both California and Texas; among the ten specification tests, the two smallest p-values are 0.071 and 0.150.[63] In Tables 3 through 6, we augmented our presentation

[61] These are distributed F with one numerator degree of freedom and a large denominator degree of freedom under the null of smoothness.

[62] Maternal education was not included on the California and Texas birth certificates until 1989, so we are unable to conduct the specification test for that variable.

[63] Note that there is no reason to expect equality of the two estimates of the discontinuity in fraction black: the one is the measured at birth, and the other is measured at motherhood.

of IV/RD estimates with IV/DOB estimates. These latter estimates invoked the original Angrist and Krueger (1991) assumption that individuals born in different times of the year are exchangeable. It is interesting to note that the global polynomial regression we used to test the regression discontinuity assumptions may also be used to test the more restrictive assumptions which would justify the IV/DOB estimates. We are particularly interested in such tests, in light of the statistical similarity of the two estimates in the empirical analysis above. A test of exchangeability may here be usefully thought of as a test for *flatness* of predetermined characteristics in birthday. We thus may easily test the Angrist and Krueger assumptions with an F-test for the exclusion of the entire global polynomial model. These statistics are presented in the row below the first F-statistic, with p-values in brackets given beneath.[64] Scanning across the columns, we see that the null hypothesis of no seasonality of birth is rejected decisively for all maternal characteristics, with all p-values below 0.001.

The second block of the table presents further evidence regarding smoothness of predetermined characteristics, but for characteristics of the grandmother. The grandmaternal characteristics we consider are fraction black, age at birth, fraction born in the U.S., birth order, and the fraction of the grandmother's live-born children who were still living at the time of the mother's birth (based on the grandmother's self-report). As with the analysis above, there is little evidence of discontinuity. Among the ten specification tests, the two smallest p-values are 0.052 and 0.158. As with the F-statistics for maternal characteristics, we see that grandmaternal characteristics are highly seasonal, implying a rejection of the identifying assumptions underlying the IV/DOB estimates. The final block of the table gives age of grandfather at delivery of the baby. For California, there is little evidence of any discontinuity (p-value of 0.105). For Texas, there is some marginal evidence of a negative discontinuity (children born just after September 1st have slightly younger fathers than children born just before). However, we hesitate to emphasize this result, since one rejection among 22 specification tests is unsurprising, given a test of 5% size. Consistent with the results in the rest of the table, we reject the null hypothesis of no seasonality in grandpaternal age.

The decisive rejections of the identifying assumptions of the IV/DOB estimates is somewhat surprising, since the IV/DOB estimates are not statistically distinct from the IV/RD estimates. We conclude that the confounders we examine here either do not matter for the outcomes we study, or are countervailing and offset one another.

IV.G. Age-for-Grade Effects

Our research design implicitly compares women who are old for their grade (i.e., those born immediately following the school entry cutoff date) with women who are young for

[64] Under the null hypothesis of flatness, these are distributed F with seven numerator degrees of freedom and a large denominator degrees of freedom.

their grade (i.e., those born immediately preceding the date). A child's relative maturity arguably could affect academic performance and behaviors, and thereby, potentially explain why we find no effects of education on fertility or infant health.

While we cannot entirely rule out this hypothesis, we argue against it on several grounds. First, the direction of bias due to an old-for-grade effect, at least for fertility, is theoretically ambiguous. On the one hand, a woman old for her grade may be mature and thus, able to fend off advances of persuasive boys. But being old for one's grade could make pregnancy more likely as older females may have already reached menarche. Second, existing evidence on the effects of age-for-grade are often detrimental, small in magnitude, and declining with age (see Stipek (1992) for a review of this literature).[65] Using a similar research design and data from a large school district in the U.S., Matsudaira and McCrary (2004) compare individuals who were born before and after the school entry dates. They estimate no differences in grade repetition for children born just prior and just subsequent the school entry date, suggesting that academic performance is unaffected by age for grade.

While we find these two lines of reasoning compelling, we can actually use our data to test whether such age-for-grade effects confound our estimates. First, we can take advantage of changes in the school entry dates across different states over time using recently-acquired national natality data with exact maternal date of birth. A change in the school entry date alters an individual's age relative to his or her peers. For example, suppose a state moves the entry date from December 1st to September 1st. A female born on August 1st before the date change will be four months older than the youngest in her cohort, but after the change, she would be one of the very youngest. A comparison like this is informative regarding age for grade if adjacent cohorts in the same state are exchangeable, a stronger assumption than we have previously considered.

In Table 8, we implement this idea and test whether the probability of motherhood responds to a change in the school entry cutoff date.[66] In particular, we estimate whether cohorts following the date change, who became younger for their grade, had different fertility behaviors when compared to earlier cohorts.[67] We find no statistical or economically-meaningful differences in the probability of giving birth, suggesting a minimal role of age-for-grade effects in explaining our fertility results.

As a final method of assessing the importance of the age-for-grade effects, we examine the reduced-form relationships by maternal age in Figure 13. From earlier estimates in Figure

[65] Traditionally, age-for-grade effects have mainly interested education researchers. However, a recent literature on this topic has sprung up in economics. Some recent studies include Datar (2005) and Bedard and Dhuey (2005), who do find age-for-grade effects on educational outcomes.

[66] We have chosen the five biggest states with the most significant changes in their school entry cutoff dates.

[67] For each date change, we identify individuals born up to three years preceding or up to three years following the cutoff date change. If the change was phased in (e.g., the date altered by a month for several years), we considered the last date change as the main policy change.

3, we know that the school entry policies do not manipulate education for older, first-time mothers. Therefore, any observed differences by day of birth in infant health outcomes are arguably age-for-grade effects.[68] Moreover, under the strong assumption of homogenous effects of education across age, the reduced-form estimates should exhibit the same patterns across age as do the first-stage estimates. The reduced-form estimates for the oldest women are never statistically significant (standard errors not shown), and overall, these estimates oscillate around zero. Together, these results suggest that relative maturity has a minimal effect on fertility decisions and subsequent infant health.

V. Discussion

Our results suggest that increases in maternal education lead to no improvements in infant health or changes in fertility behaviors, a great contrast with the recent findings of Currie and Moretti (2003), arguably one of the more comparable studies to our own. Using openings of two- and four-year institutions in a woman's county at age 17 as an instrument for education, Currie and Moretti conclude that maternal education has a substantial influence on infant health and fertility patterns.[69] For both prematurity and low birth weight, they find that a one year rise in education reduces the incidence of these outcomes by a sizable one percentage point. For our more precise specifications, these effect sizes are outside of the confidence intervals for our estimates.

This discrepancy highlights the plausible heterogeneity in the causal effect of education. Education can affect health and fertility through several different channels, the mostly frequently mentioned of which are income and health knowledge. One view is that our estimates reflect only the effect of education through learning. The group of mothers of Currie and Moretti's study, women aged 24 to 45, are old enough to have experienced the labor and mating market consequences of their education. In contrast, the mothers in our study may be too young to have entered the labor force yet.

On the other hand, in the absence of borrowing constraints, these young mothers should correctly anticipate the labor and mating market consequences of their education and behave accordingly. Our data do provide some suggestive evidence that young mothers respond to the permanent income effects of their education. Table 9 presents estimates of the effect of education on the payment method for delivery (California) and the source of prenatal care (Texas). For California, an extra year of education reduces the likelihood of public payment for delivery (e.g., Medicaid) and raises the likelihood of private payment (e.g., private insurance), leaving self-payment (i.e., out-of-pocket) unaffected. In most cases, the recipiency of public funding is dependent on income. This suggests that, already at motherhood, the

[68] Women who delay childbearing tend to be of high socioeconomic status. It is possible for these women, age-for-grade effects are less important than for our main sample of mothers.

[69] These estimates are specific to white mothers, as college openings do no appear to affect educational attainment for black women.

women in our study are experiencing differences in income due to their education. For Texas, an extra year of education lowers the likelihood that a woman receives prenatal care in a hospital and raises the likelihood that she receives care in a private clinic, leaving unchanged the likelihood of care in public clinic. These results may reflect anticipated longer-term income consequences of education.

Another possible explanation for the disparate findings is that college openings affect smoking rates, whereas school entry policies do not. In the model of Section II, the effect of education on health is specifically a combination of the effect of schooling on behavior and the effect of behavior on health. Even if the effect of smoking on infant health is constant, it is entirely plausible that the effect of schooling on behavior (i.e., maternal smoking) is heterogeneous. It is interesting to note that a woman with more education, by virtue of her augmented household income, may be more able to afford cigarettes. For the women in our study, it is possible that this income effect offsets the health knowledge gained in school. For the women in the study of Currie and Moretti, the health knowledge effect appears to dominate the income effect.[70]

VI. Conclusion

In this paper, we have presented evidence against an important role for education in a woman's fertility and the health of her child at birth. Our evidence is based on comparisons of several key outcomes between women born just before and after the school entry date. Compared to women born just before the school entry date, women born just after the entry date (i) have substantially lower schooling, as expected, (ii) are equally likely to become mothers, (iii) give birth at similar ages, and (iv) give birth to similarly healthy infants. That we do not document differences in infant health is surprising, given the assortative mating results: school entry policies lead to economically important differences in the age and education of a woman's mate.

These comparisons are essentially as good as comparisons based on randomization provided that confounders are smooth in day of birth. On prior grounds we find it credible that two individuals born near in time are exchangeable. To substantiate this point, we have provided evidence that measured pre-determined characteristics are nearly identical for women born just before and after the school entry date.

Our estimates are specific to the subpopulation of women whose education at motherhood is affected by school entry policies. These women may be negatively selected, for several reasons. First, their parents were willing to comply with school entry policies; anecdotal evidence suggests that higher income parents often circumvent these policies. Second, school

[70] In addition, one might imagine that there are peer and job-stress effects on smoking. Peer effects would suggest that college attendance would reduce the incidence of smoking, whereas job-stress effects would suggest that college attendance would raise the incidence of smoking.

entry policies affect education at motherhood for those women giving birth at young ages with low education generally. Thus, these results may be difficult to generalize to other subpopulations, possibly explaining the discrepancies between our estimates and those of Currie and Moretti (2003).

On the other hand, this may mean that our results are relevant for specific policies. The National Campaign to Prevent Teen Pregnancy, a non-profit and non-partisan initiative, emphasizes the importance of schooling in reducing rates of teenage pregnancy. However, our results suggest that such emphasis may be misplaced. When policymakers envision expensive interventions to raise female education, they should think carefully of how they expect increases in education to improve well-being. With teenagers, specifically, they should be cautious in their expectations of the benefits of such growth in education attainment.

Finally, these estimates directly address the fertility and infant health consequences of starting school early. Parents of children with birthdays near the school entry date may be interested in these findings, particularly if they view their child as at risk of dropping out of school. Moreover, there continues to be an active policy debate regarding the appropriate age at school entry, and several states have changed the school entry date to earlier in the year in order to raise the average age of kindergartners. Our results suggest that even if moving back the entry date does succeed in improving the preparedness of children for school, such a policy shift is likely to create both winners and losers.

Appendix I

THEOREM. *Suppose that (Y, S, R) are observed for the selected sample with $M = 1$, where the data generating process is as in system (8)-(10) with $\tau = 0$, and suppose the conditioning event $R = r, w > -\overline{M^*}$ is non-trivial. Then least squares consistently estimates π and ρ.*

PROOF. Least squares will be consistent for π provided that $E[u|R = r, w > -\overline{M^*}]$ is a continuous function of r (a proof of consistency of least squares for ρ follows the same argument). Suppose that the residuals possess a joint density function conditional on $R = r$ (a similar proof holds for discrete and mixture cases). Then note that

$$E[u|R = r, w > -c] \quad = \quad \frac{\int_{-\infty}^{\infty}\int_{-c}^{\infty} uf(u, w|r)dwdu}{\int_{-\infty}^{\infty}\int_{-c}^{\infty} f(u, w|r)dwdu} \equiv h(r, c)$$

where $f(u, w|r)$ is the joint density function for (u, w) conditional on $R = r$. Without loss of generality, suppose that $f(u, w|r)$ is continuous in r (if it were not, then the setup of the problem is such that the parameterization of the system in (8)-(10) could be altered to make it continuous). Continuity of $f(u, w|r)$ in r implies continuity of $h(r, c)$ in r since the conditioning event is assumed to be non-trivial. Moreover, $h(r, c)$ is continuous in c by Leibniz' rule since differentiability implies continuity. Then note that if $\tau = 0$, $\overline{M^*}$ is continuous in r. But then $h(r, \overline{M^*})$ is continuous in r, as required. □

Appendix II

In this appendix, we discuss details of the implementation. Section A describes the uniform confidence regions presented in Figures 3, 6, 8, and 10, while Section B discusses a statistical approach to robustness that justifies the global polynomial approximations to the conditional expectations we estimate in this paper.

A.II.A. Uniform Confidence Regions

Several of the figures we present in Section IV give estimates of the *distribution* of treatment effects. The twice standard error bands shown in these figures are based on uniform confidence region ideas and are calculated according to a parametric bootstrap procedure, which we describe below in the form of an algorithm, in the specific context of Figure 4 for concreteness.

Before turning to the algorithm, recall that discontinuity estimates such as those in Table 2 can be estimated either directly from the micro-data, or from local averages, and that these two approaches are asymptotically equivalent. To make the algorithm clear, it helps to define some notation. For schooling level s, let $\widehat{\beta}_{s,r}$ denote the fixed effect for the rth

day of birth from a regression of an indicator for whether an individual has years of schooling s or less on the design matrix $\mathbf{W}$ (containing the indicators for each day of birth) and $\mathbf{Z}$ (containing individual level controls but not including a constant column). Let $\mathbf{X}$ denote the design matrix for the polynomial model for the conditional expectation, as in equation (12), but for the group-level regression. For clarification of the notation, observe that $\mathbf{W}$ is a $n \times J$ matrix, where J is the number of groups (here, $J = 366$), $\mathbf{Z}$ is a $n \times K$ matrix, where K is the number of individual level controls, and $\mathbf{X}$ is a $J \times M$ matrix where M is the number of coefficients in the polynomial model in equation (12). [71] Let $\widehat{\beta}_s = (\widehat{\beta}_{s,r})_{r=-182}^{183}$ denote the column vector of the fixed effects for each day of birth, and let $\widehat{\Omega}_s$ denote the estimated variance matrix for $\widehat{\beta}_s$ (in the absence of controls, $\widehat{\Omega}_s$ will be diagonal). Then let $\widehat{\eta}_s$ denote the discontinuity estimate for the probability of having schooling of s years or less (i.e., the element of $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\widehat{\beta}_s$ corresponding to D, as in equation (12)), and let $\widehat{\sigma}_{\eta_s}$ denote the standard error for discontinuity estimate s. We are now in a position to describe the

Uniform Confidence Region Algorithm:

1. For bootstrap draws $b = 1, 2, \ldots, B$, draw a 366-vector, $\widehat{\beta}_{s,b} = (\widehat{\beta}_{s,b,r})_{r=-182}^{183}$, from the multivariate normal distribution with mean of $\widehat{\beta}_s$ and variance of $\widehat{\Omega}_s$. For each bootstrap draw b, compute $\widehat{\eta}_{s,b}$, the element of $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\widehat{\beta}_{s,b}$ corresponding to D.

2. Collect the estimates $\widehat{\eta}_{s,b}$ into a data set with B rows and K columns, where K is the dimension of the support of schooling. For each element of the data set form $\widehat{\kappa}_{s,b} = \left|(\widehat{\eta}_s - \widehat{\eta}_{s,b})/\widehat{\sigma}_{\eta_s}\right|$ or the absolute value of the t-test statistic for whether $\widehat{\eta}_{s,b}$ is in the confidence region for η based on the original data set.

3. For each schooling level s, calculate the 95th percentile of $\widehat{\kappa}_{s,b}$ (over the $b = 1, 2, \ldots, B$ observations) and verify that it is roughly within sampling variation of 1.96. (In a similar spirit, one may verify that the indicator for the event $\widehat{\kappa}_{s,b} > 1.96$ has an empirical probability of success within sampling variation of 95%.)

4. Calculate $\widehat{\kappa}_b = \max_s \widehat{\kappa}_{s,b}$ (i.e., add an additional column to the data set), and compute the 95th percentile of $\widehat{\kappa}_b$, labeling this latter quantity $\widehat{T}$.

5. Form confidence regions $\widehat{\eta}_s \pm \widehat{T}\widehat{\sigma}_\eta$.

[71] Sphericality of the errors is not necessary, but we find it perfectly palatable in this context. Note that it is partially testable (see Step 3 of the algorithm). Note also that in other contexts, alternative variance assumptions could be imposed, such as clustering at the county-day of birth level. However, we caution researchers to notice that clustering at the day of birth of level is inappropriate and guaranteed to yield $\widehat{\sigma}_{s,r} = 0$ for every r and s.

The key idea of the algorithm is the final step. Intuitively, the algorithm replaces the conventional critical value of 1.96 with the more conservative and appropriate critical value, $\widehat{T}$, estimated from a bootstrap procedure. The algorithm corrects for the "drift" away from 5% in the probability of obtaining estimates statistically significantly different from a true null hypothesis, as the number of hypotheses being tested increases. For example, suppose that there was no effect of school entry policies on the distribution of age at first birth. Because the support of age at first birth is roughly fifteen to forty-five, we test some thirty hypotheses simultaneously. It is not hard to see that this leads to an elevated probability of rejecting one of these thirty hypotheses—arguendo, if the tests were taken to be independent then a simple exercise in probability shows that the appropriate critical value is roughly 3.14, which is substantially above 1.96. More generally, the appropriate critical value will depend on the degree of dependence across the tests. The complexity of this dependence in the presence of control variables is a powerful argument for replacing analytical calculations with bootstrap-type procedures.

A similar procedure is outlined in Chernozhukov (2002) and Angrist, Chernozhukov and Fernandez-Val (2004).[72] The difference between our procedure and theirs is that we replace subsampling with a parametric bootstrap procedure. The parametric bootstrap is a computationally convenient way to approximate a $\binom{n}{n}$ resampling from the original micro-data and ensures that data near the discontinuity are available for each bootstrap draw (this may be a problem with subsampling).

A.II.B. Local Averages and Robustness

Estimation of the parameters of a regression discontinuity model requires that the researcher make subjective smoothing choices, the validity of which may be disputed by other researchers. For example, in Figure 1, we presume that the conditional expectation of schooling given day of birth may be summarized using 8 parameters. However, the chosen functional form may be over- or under-parameterized or might not even be among the class of functions in which the true conditional expectation falls. Because of these ambiguities, it is important to reveal the sensitivity of the parametric results to different smoothing and estimation choices.

This is an important role of the local averages presented throughout this paper. To describe the calculation of these, consider Figure 1, where the local averages are presented as open circles. Each local average in the figure is the conditional mean of education at motherhood for a separate day of birth, where the conditioning is based on the same set of control variables used for the parametric model: (i) indicators for race, (ii) indicators for ethnicity, (iii) indicators for age, and (iv) indicators for year of data. Practically, these conditional means are obtained as the fixed effect coefficients from a regression of schooling

[72] Both the referenced articles contain proofs and references to the relevant literature.

on a set of dummies for the day of birth and a saturated model for the control variables.[73] As is widely understood, estimates for a regression model with a grouping structure may either be obtained directly from a micro-level regression or indirectly from a weighted regression at the group level—that is, these estimators are numerically identical.When there are micro-level control variables, as in our implementation, the micro-level regression estimates and the group-level regression estimates are nearly the same, but the fixed effect coefficients from a first-step regression replace the group means, and first-order asymptotic equivalence replaces numeric equality (the asymptotics invoked here assume that the number of groups and the number of observations within group are both large).

This result is important because it means that, to first order, any interesting estimator for the discontinuity in maternal schooling at the school entry date may be understood as a smoothed version of the local averages presented in the figure.

A more formal manner of assessing the validity of the restrictions imposed by the parametric model is also available. The basic idea of the test is that if the parametric model is correct, then the average sampling variance of the maternal schooling conditional means should be close to the mean squared error of the cell-level regression (cf. Lee and Card 2004). If there are no micro-level control variables, then the precise test statistic is

$$\widehat{F} = \frac{\sum_{j=1}^{J} \frac{n_j}{n} \left(\overline{S}_j - \widehat{S}_j\right)^2 / \left(J - M\right)}{\sum_{j=1}^{J} \frac{n_j - 1}{n} \widehat{\sigma}_j^2 / \left(n - J\right)}$$

which is distributed F with $J - M$ numerator degrees of freedom and $n - J$ denominator degrees of freedom under the null hypothesis that the polynomial model is correct, where J is the number of parameters in the less restricted model (here, the 366 possible days of birth), M is the number of parameters in the more restricted model (here, the 7 parameters of the polynomial: a constant, 2 cubics, and an intercept shifter at the entry date), $\widehat{S}_j$ are the predictions from the estimated polynomial model, and $\widehat{\sigma}_j^2 = \frac{1}{n_j - 1} \sum_{i=1}^{n_j} (S_{ij} - \overline{S}_j)^2$ is the estimated variance of schooling for the jth group. Abstracting from the factor $(n - J)/(J - M)$, the numerator is the weighted mean squared error from the group-level regression, while the denominator is a weighted average of the group-level variances.

Table A2 gives these F-statistics for all conditional expectations estimated for the reduced form analysis of Table 2. We see that of the 28 statistics, six exceed the 5% critical value of 1.13 and three exceed the 1% critical value of 1.18. These are generally corroborative of the parametric models we have imposed, but there are counterexamples.

[73]For presentation purposes, we have added back the overall sample mean of schooling to each of the fixed effect coefficients. This is necessary since we use a saturated model for the control variables. An alternative procedure that would preserve the scaling of the coefficients is to parameterize the control variables in such a way as to leave one degree of freedom free. Our procedure is a computational expedient that takes advantage of statistical software conventions. Note that this choice does not affect the point estimate of the discontinuity.

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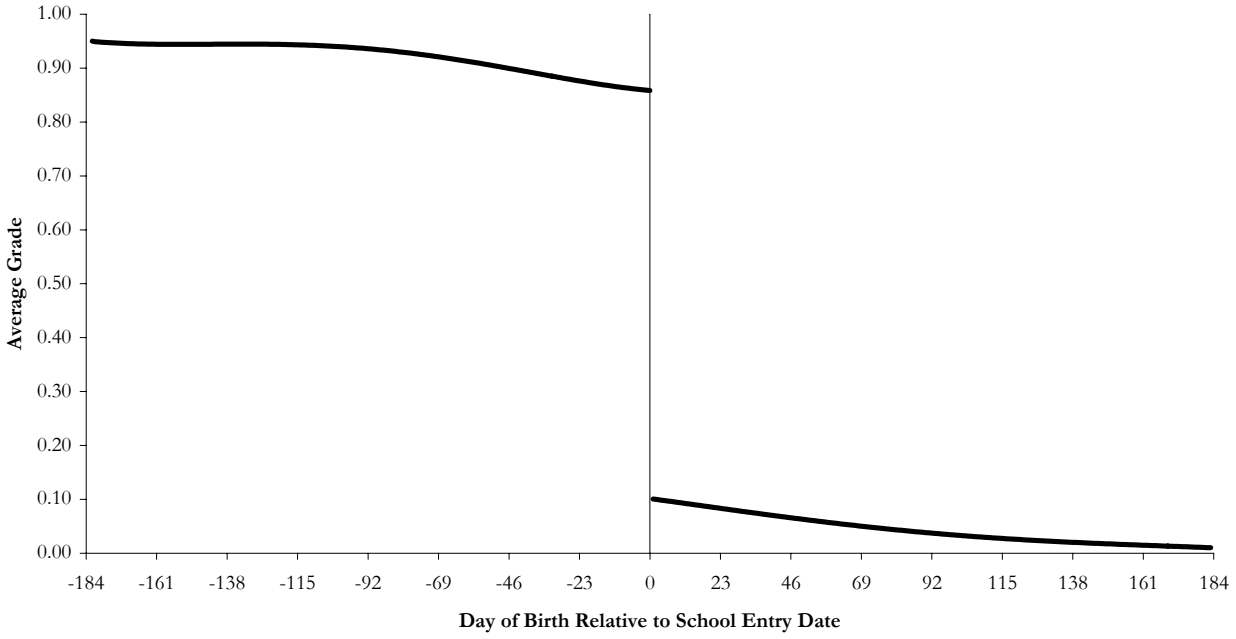
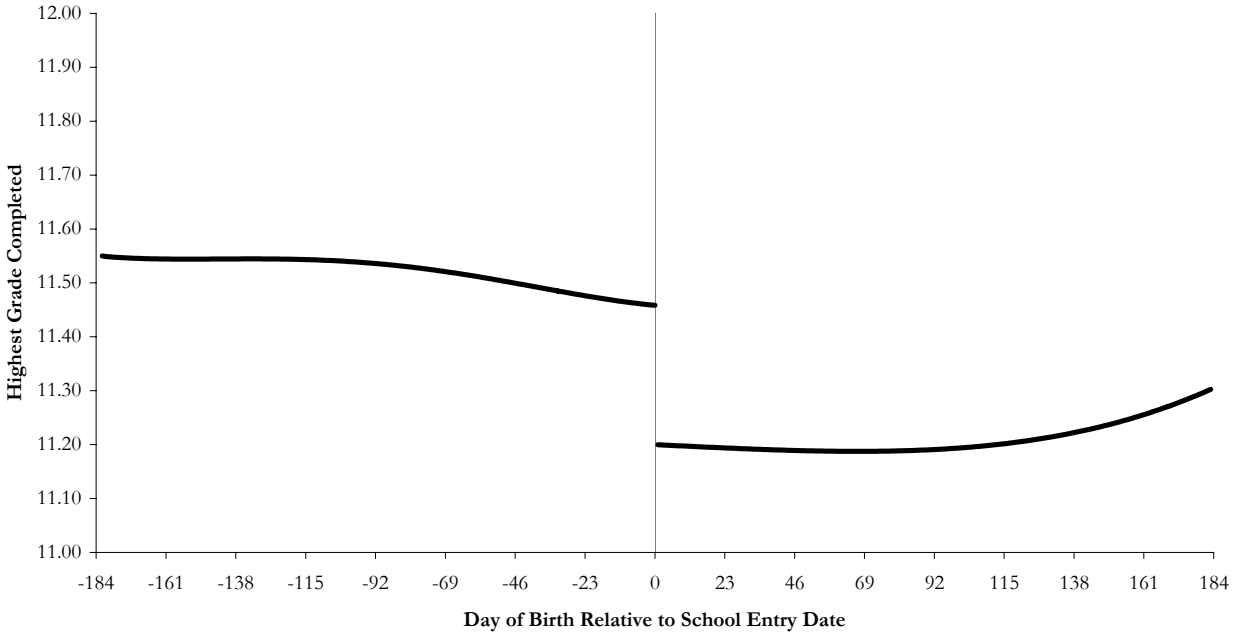
Figure 1. Schematic View of Identification Strategy:
Discontinuities in Educational Progression Created by School Entry Policies

A. Grade in School at Age Six



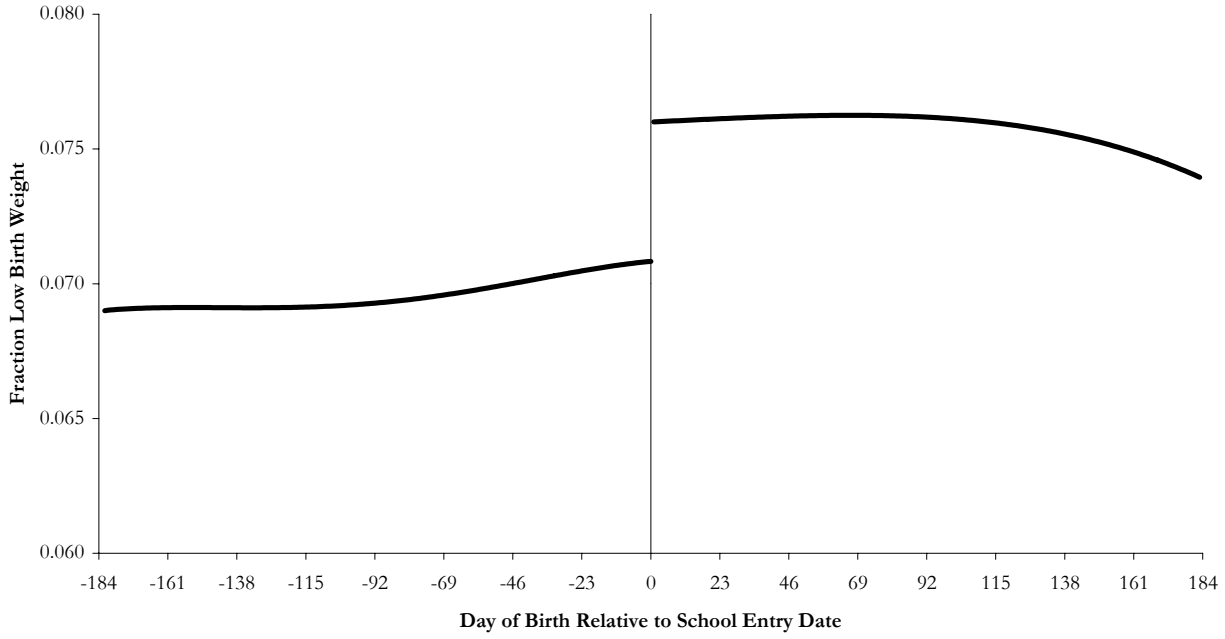
B. Education at Motherhood



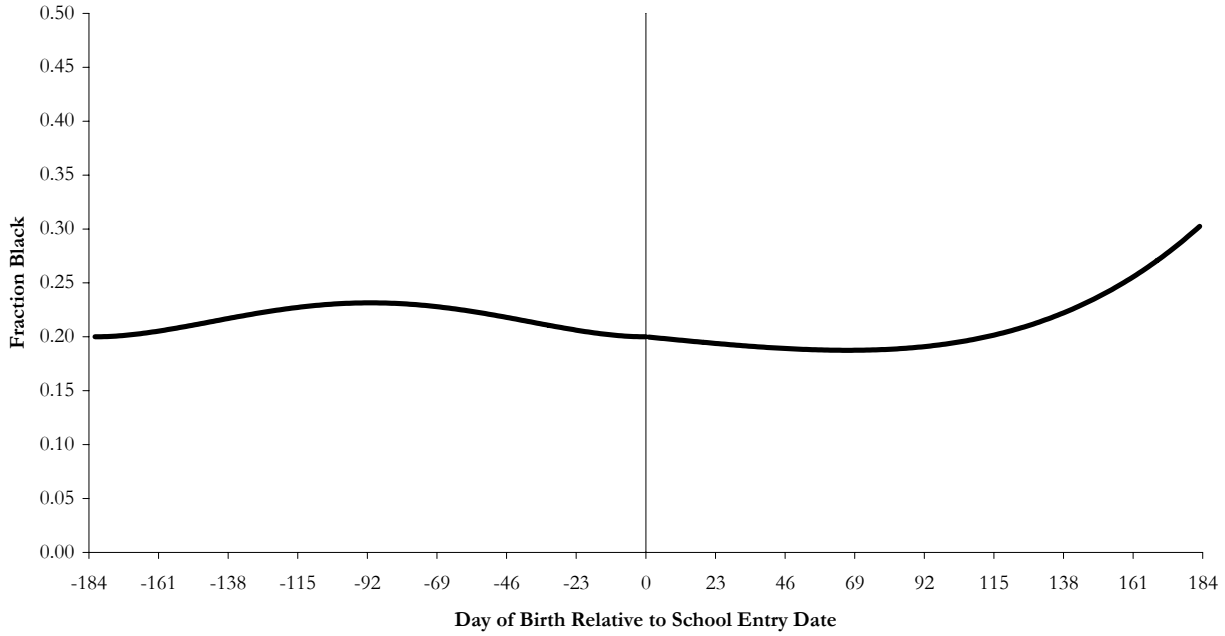
Notes: Figure gives hypothetical day of birth patterns for the outcomes specified in each panel. Panel A gives grade in school, measured as of June 30 of the year in which a cohort of children would have begun school, had their birthday fallen prior to the school entry date. Panel B gives education at motherhood. Panel C (next page) gives infant health, while Panel D (next page) gives fraction black. See text for discussion.

Figure 1 (continued). Schematic View of Identification Strategy: Predictions for Infant Health and Pre-determined Characteristics

C. Infant Health: Fraction Low Birth Weight



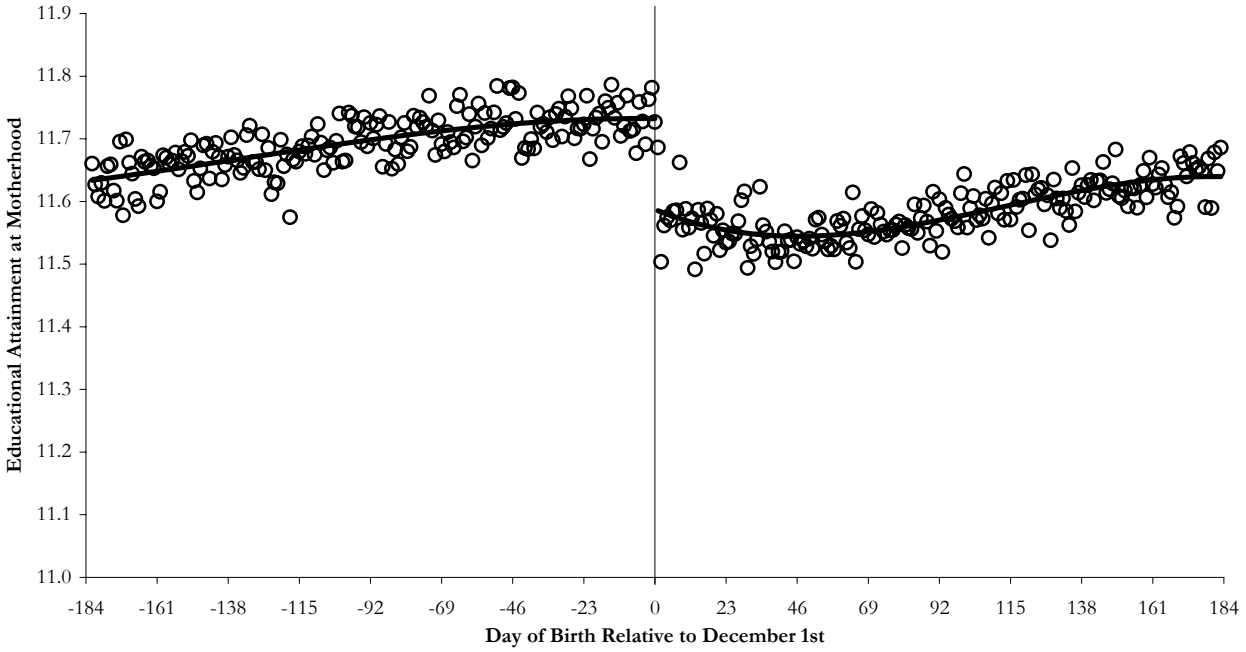
D. Pre-Determined Characteristic: Fraction Black



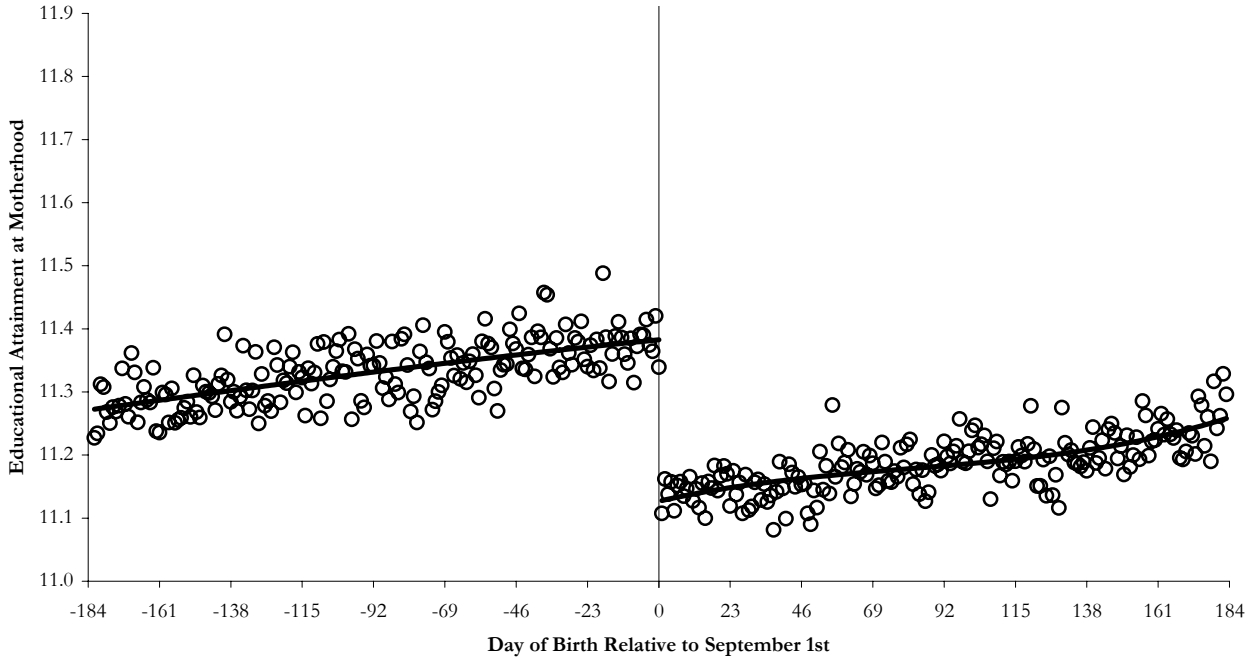
Notes: See Panels A and B (preceding page).

Figure 2. Maternal Education, by Mother's Day of Birth: Native First-Time Mothers 23 Years Old and Younger

A. California



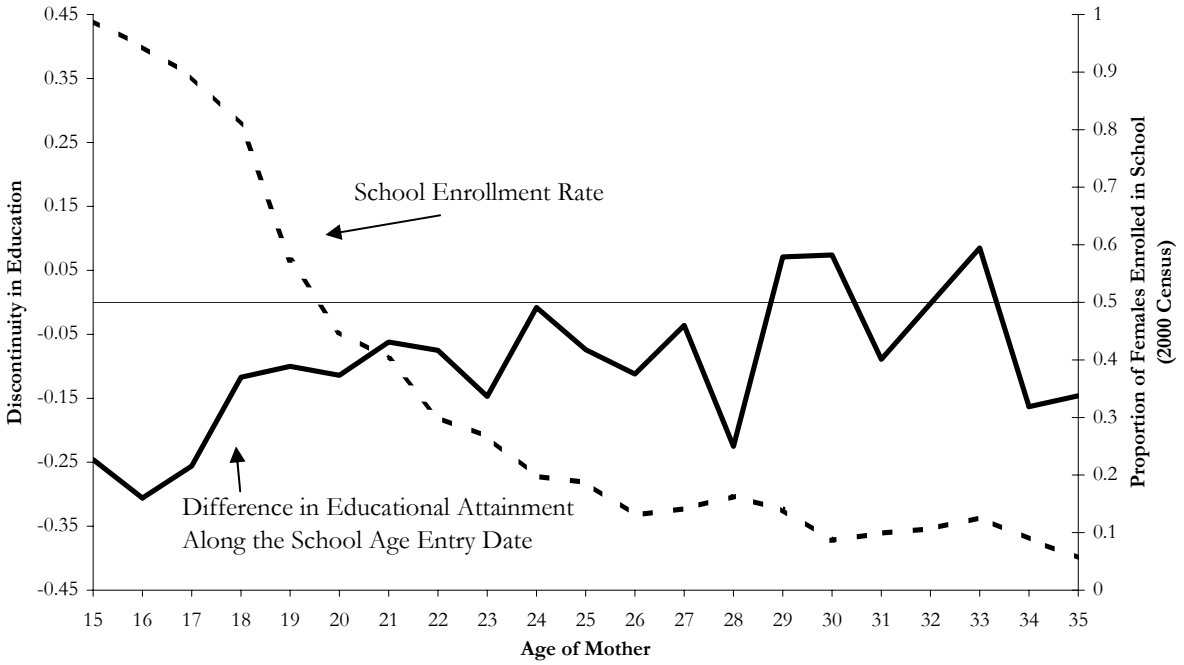
B. Texas



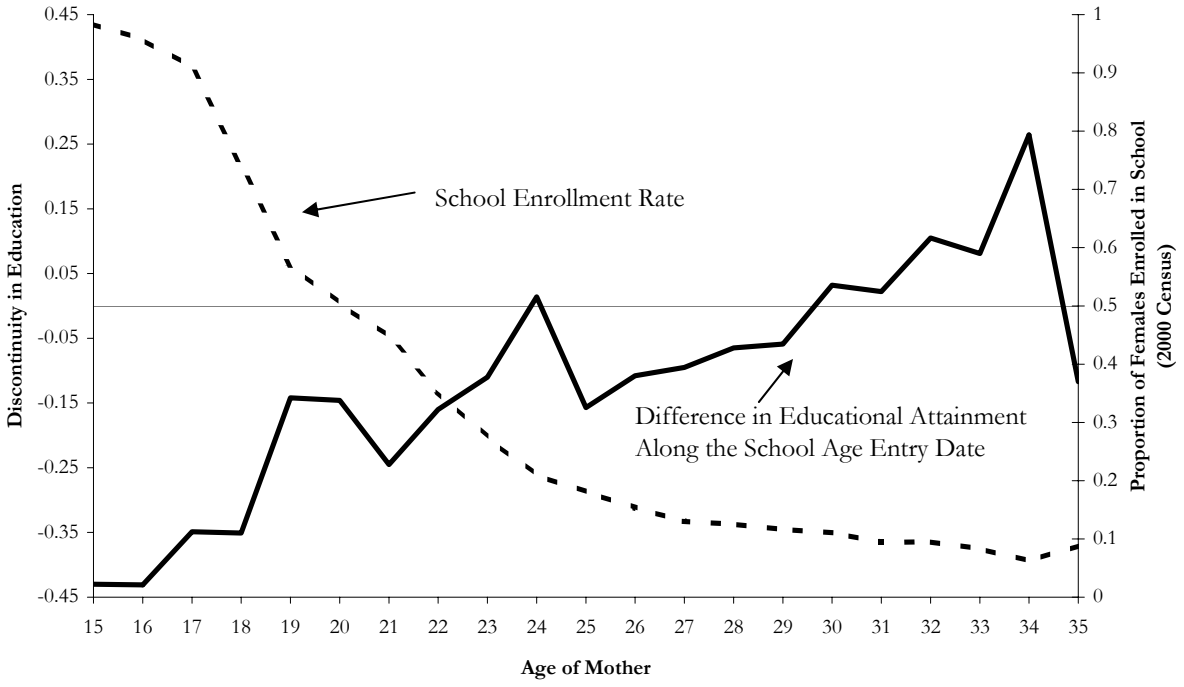
Notes: The open circles represent conditional means of maternal education by mother's day of birth (relative to the school age entry cutoff date). The controls are fixed effects (uninteracted) for mother's ethnicity/race, mother's age, and year of infant's birth. The solid lines represent an estimate of a third degree polynomial which approximates the relationship between day of birth and maternal education. See equation (6). The estimation sample includes only first-time mothers born in the state in which they gave birth who are 23 years old or younger and had singleton births.

Figure 3. Education Discontinuities, by Age of Mother

A. California

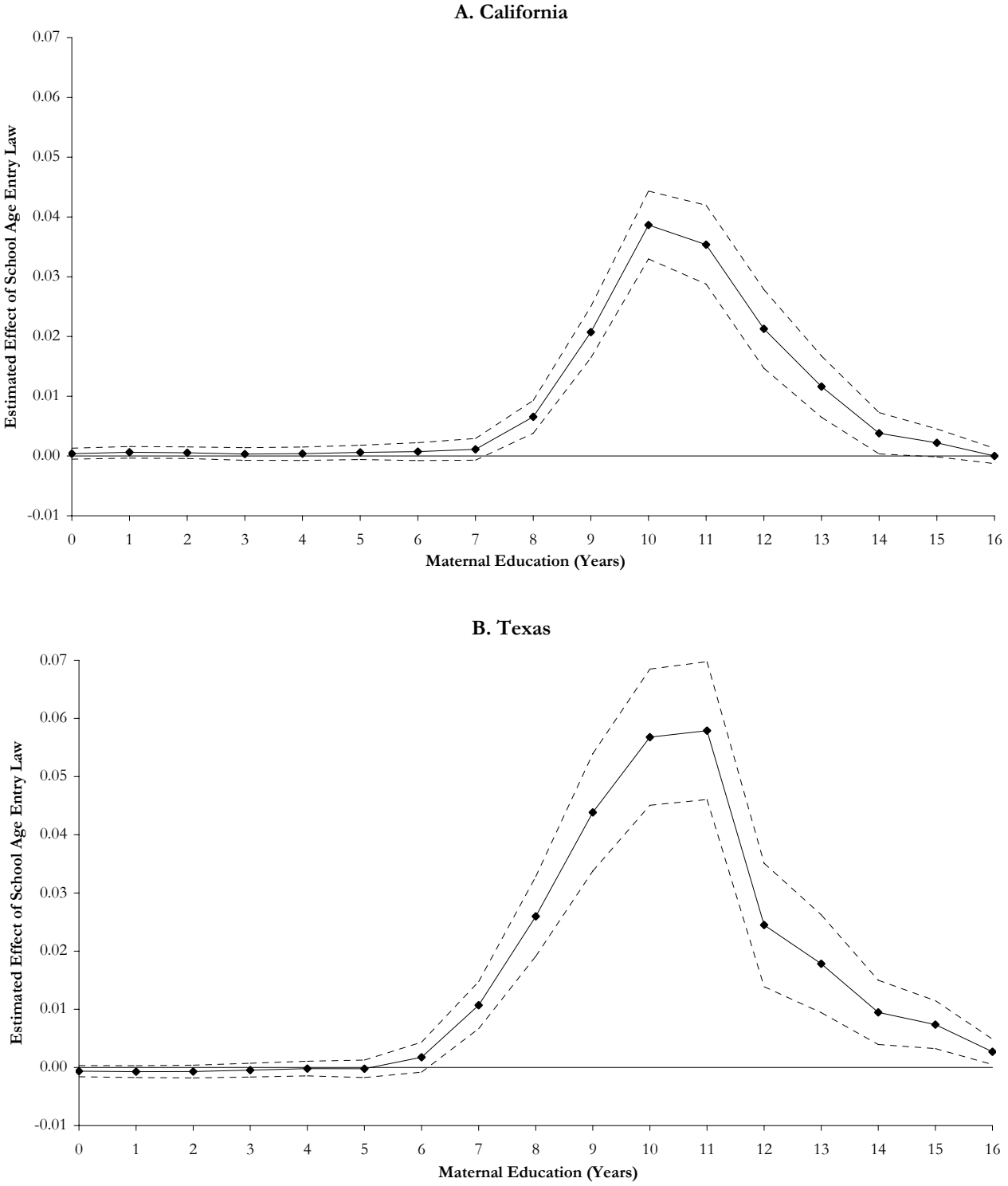


B. Texas



Notes: The solid lines represent the estimated educational difference at the cutoff dates for the sample of mothers defined by the age on the x-axis. The educational difference comes from an estimate of equation (6). The dashed lines represent the fraction of females enrolled in school by age from the 2000 Census.

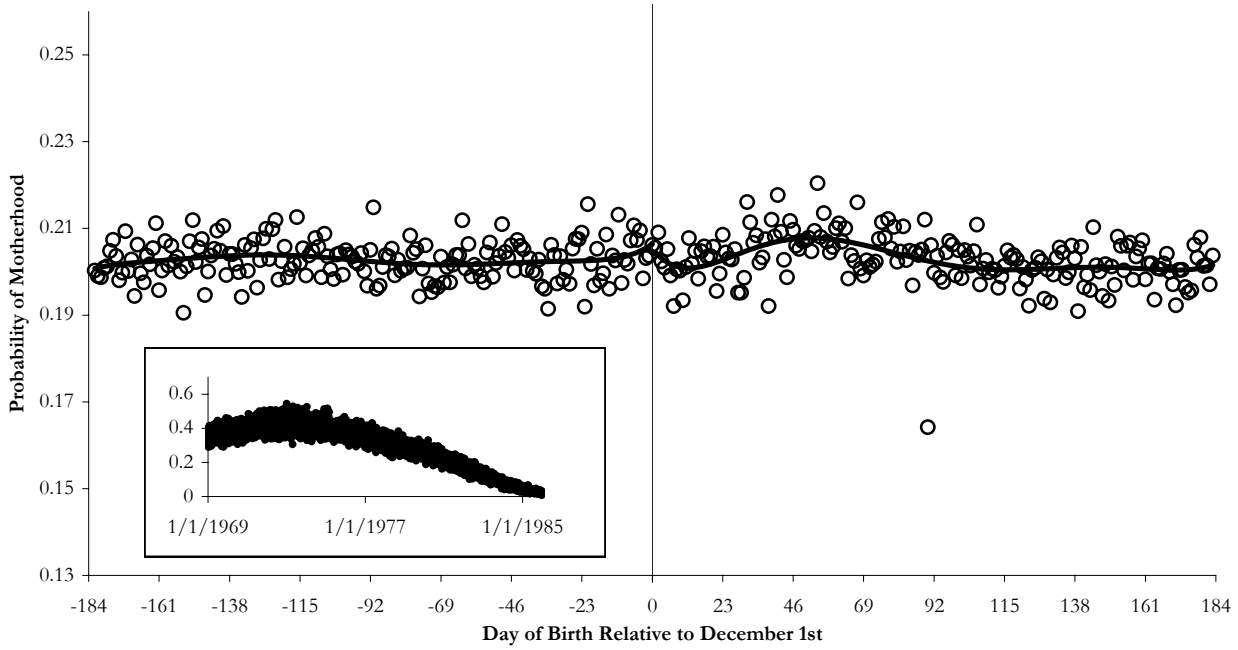
Figure 4. Distribution of Estimated Maternal Education Effects



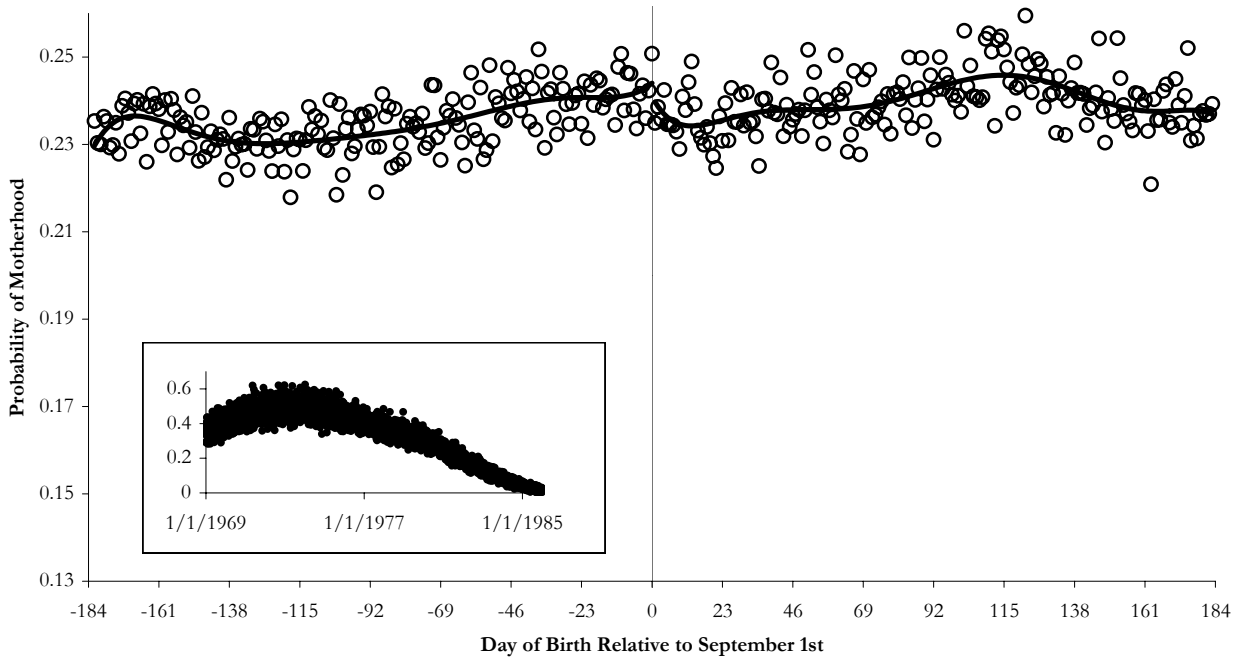
Notes: For details on the estimation sample, see Figure 1. Each point on the solid line represents the estimated difference in the probability of having less than or equal to the number of years of schooling represented on the horizontal axis between the sample of mothers born before and after the school start-age entry date. This difference in probability comes from an estimate of equation (12). The dashed lines represent the upper and lower 95% uniform confidence interval bands for the estimates represented by the solid lines. For more details on the construction of the uniform confidence intervals, see Appendix II.

Figure 5. Probability of Motherhood, by Mother's Day of Birth: Native Mothers

A. California



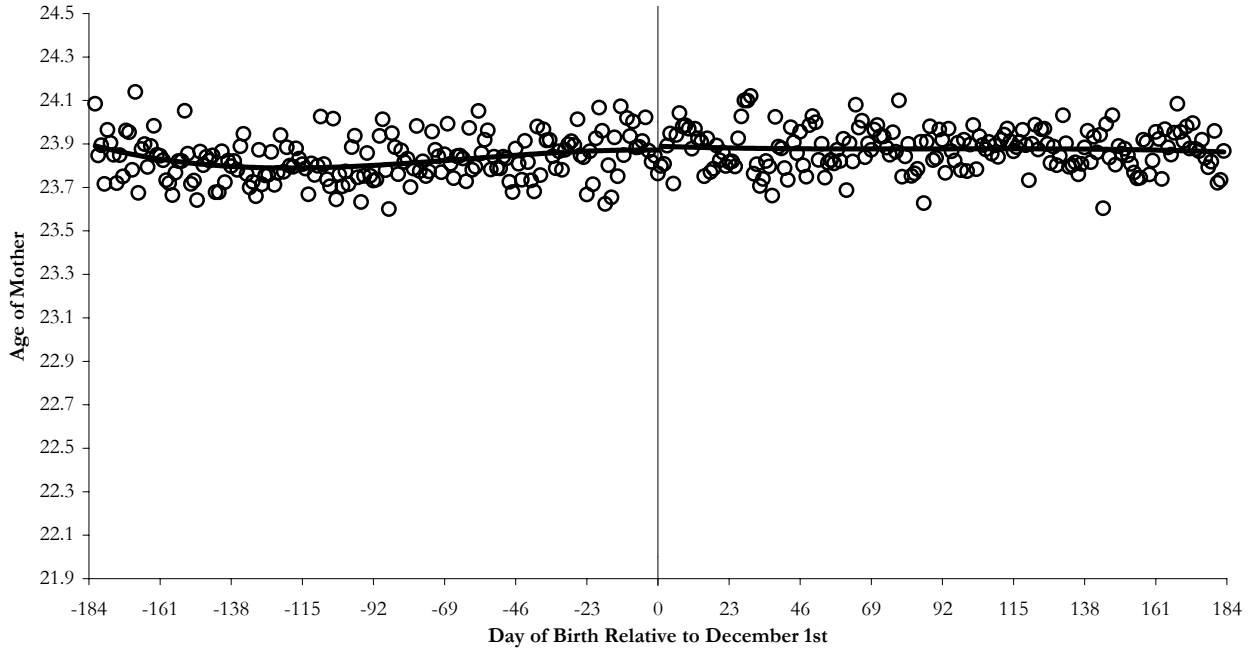
B. Texas



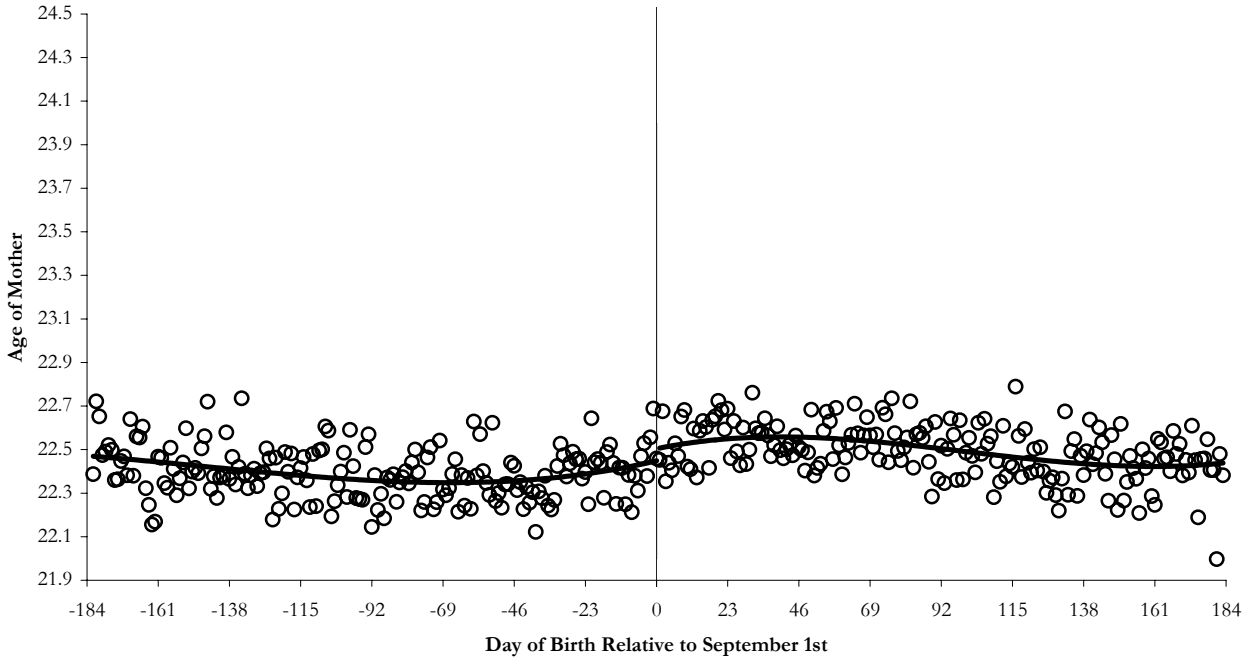
Notes: In the main part of each figure, open circles represent an estimate of the probability of motherhood by mother's day of birth relative to the school age entry cutoff date (for details, see text). The solid line gives an estimate of the probability of motherhood in mother's birthday. The estimate is based on a global polynomial model using a seventh degree polynomial in mother's birthday, an indicator for being born after the school start-age entry date, and an interaction of a seventh degree polynomial in birthday and the post-cutoff dummy to predict motherhood. The inset graph gives an estimate of the probability of motherhood, but for each possible date of birth.

Figure 6. Maternal Age, by Mother's Day of Birth: Native First-Time Mothers

A. California

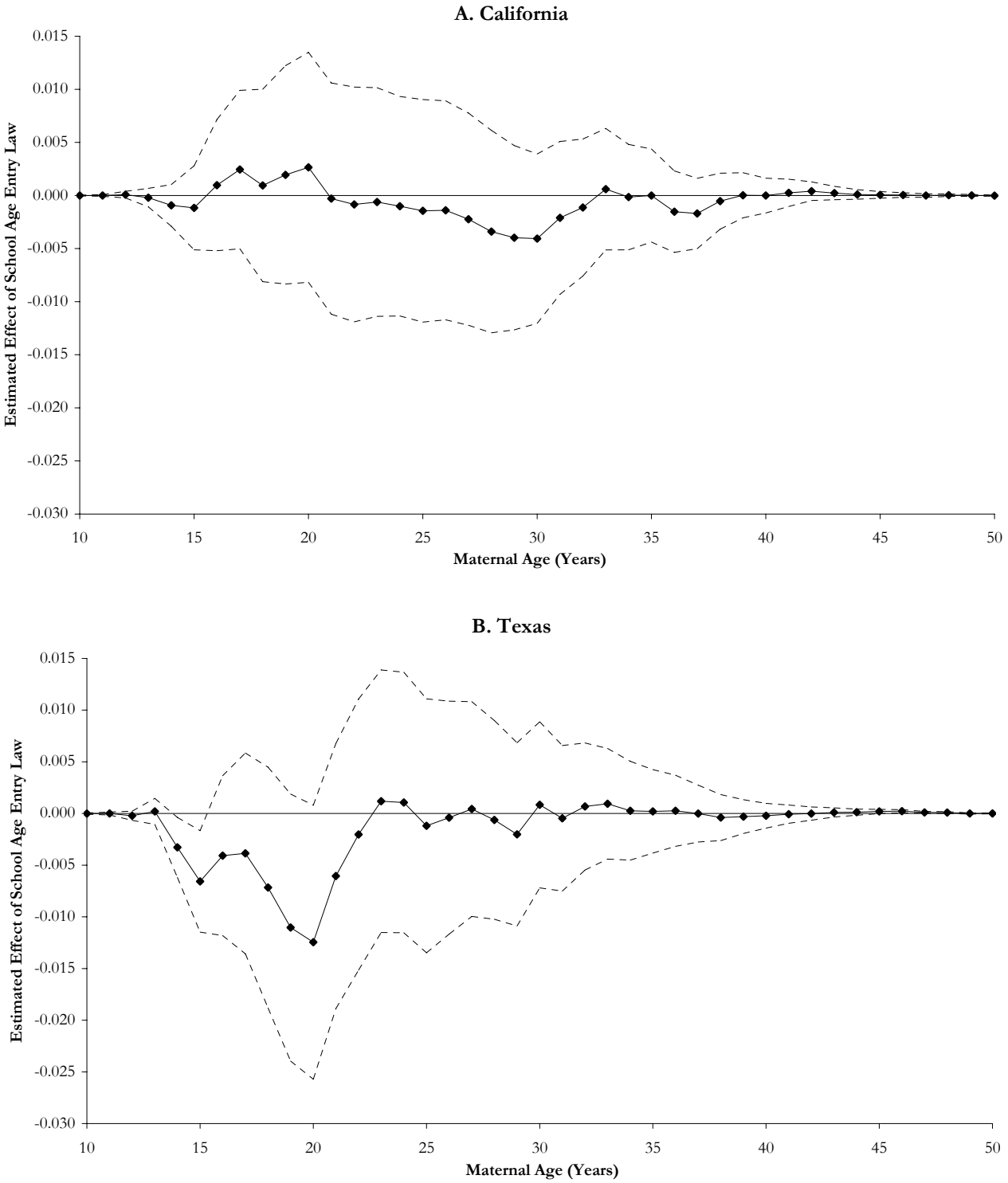


B. Texas



Notes: The open circles represent conditional means of maternal age by mother's day of birth (relative to the school age entry cutoff date). The controls include fixed effects (uninteracted) for mother's ethnicity/race and year of infant's birth. The solid lines represent an estimate of a third degree polynomial which approximates the relationship between day of birth and maternal age (i.e., an estimate of equation (6) with age replacing education as the dependent variable). The estimation sample includes only first-time mothers born in the state in which they gave birth and had singleton births.

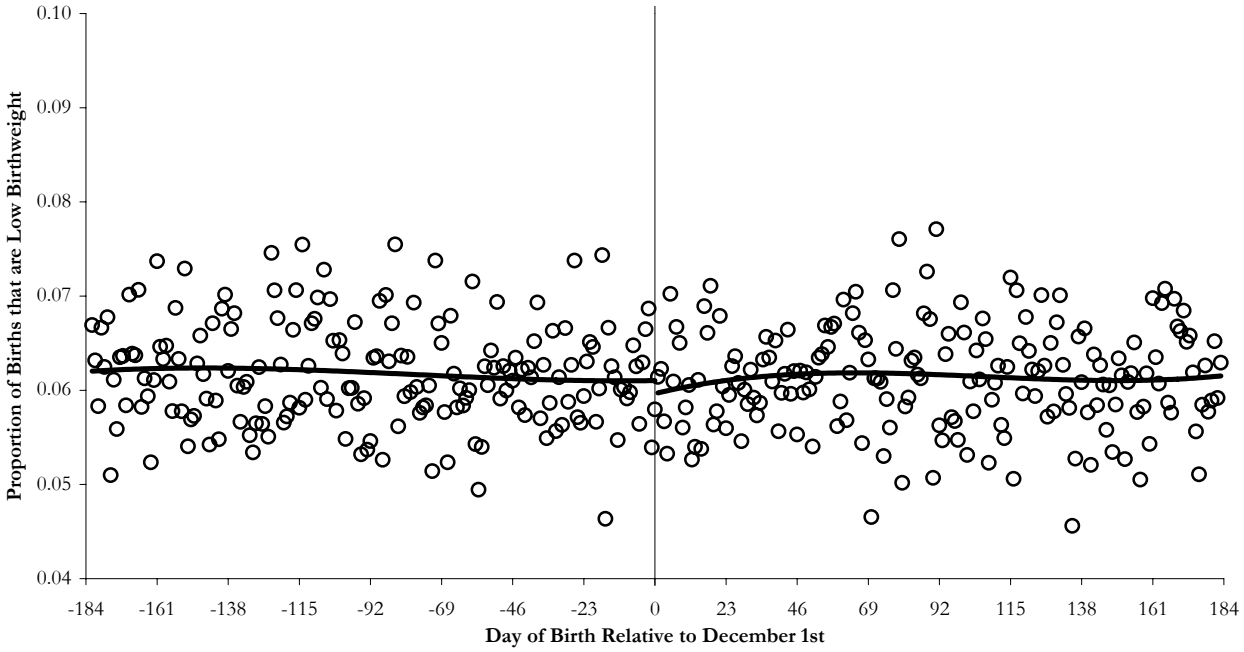
Figure 7. Distribution of Estimated Maternal Age Effects



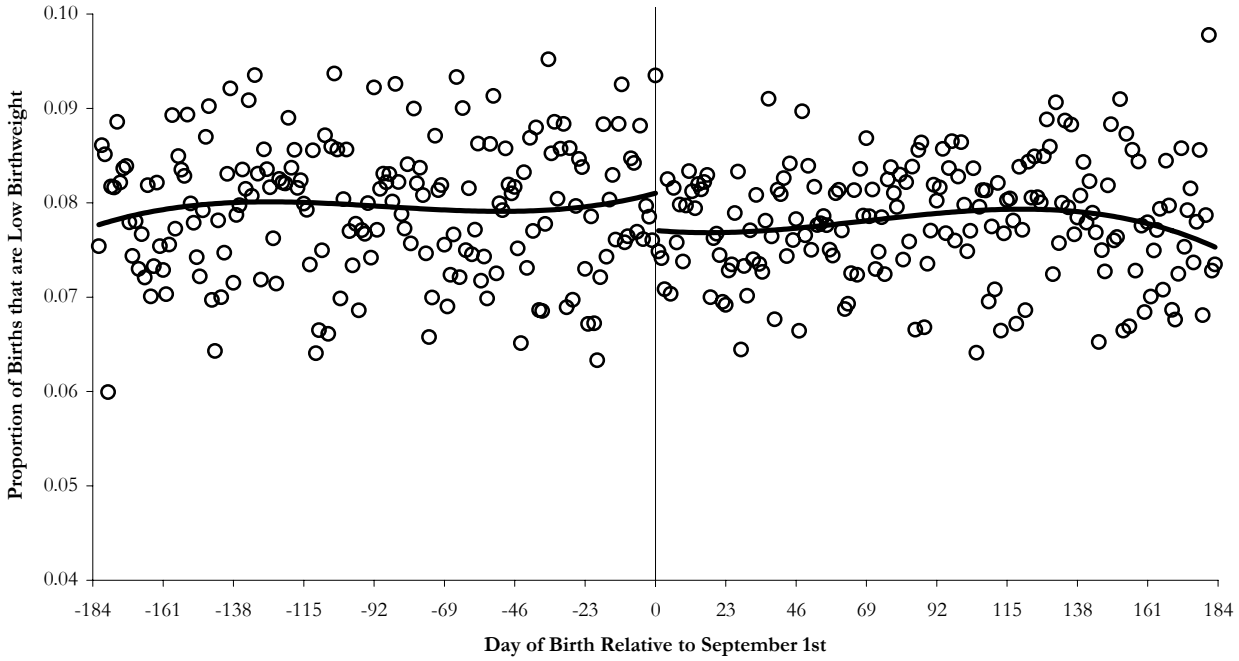
Notes: For details on the estimation sample, see Figure 5. Each point on the solid line represents the estimated difference in the probability of giving birth by the age represented on the horizontal axis between the sample of mothers born before and after the school start-age entry date. This difference in probability comes 'from an estimate of equation (12) with age replacing schooling as the dependent variable. The dashed lines represent the upper and lower 95% uniform confidence interval bands for the estimates represented by the solid lines. For more details on the construction of the uniform confidence intervals, see Appendix II.

Figure 8. Incidence of Low Birthweight, by Mother's Day of Birth: Native First-Time Mothers 23 Years Old and Younger

A. California



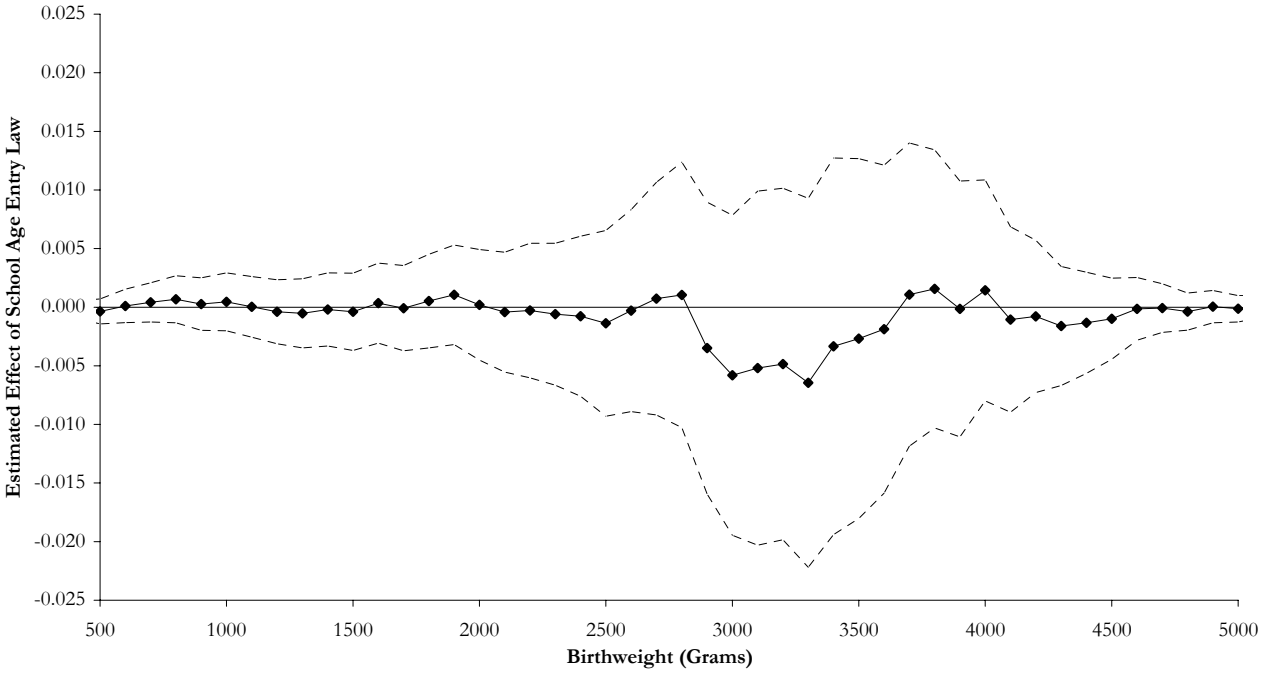
B. Texas



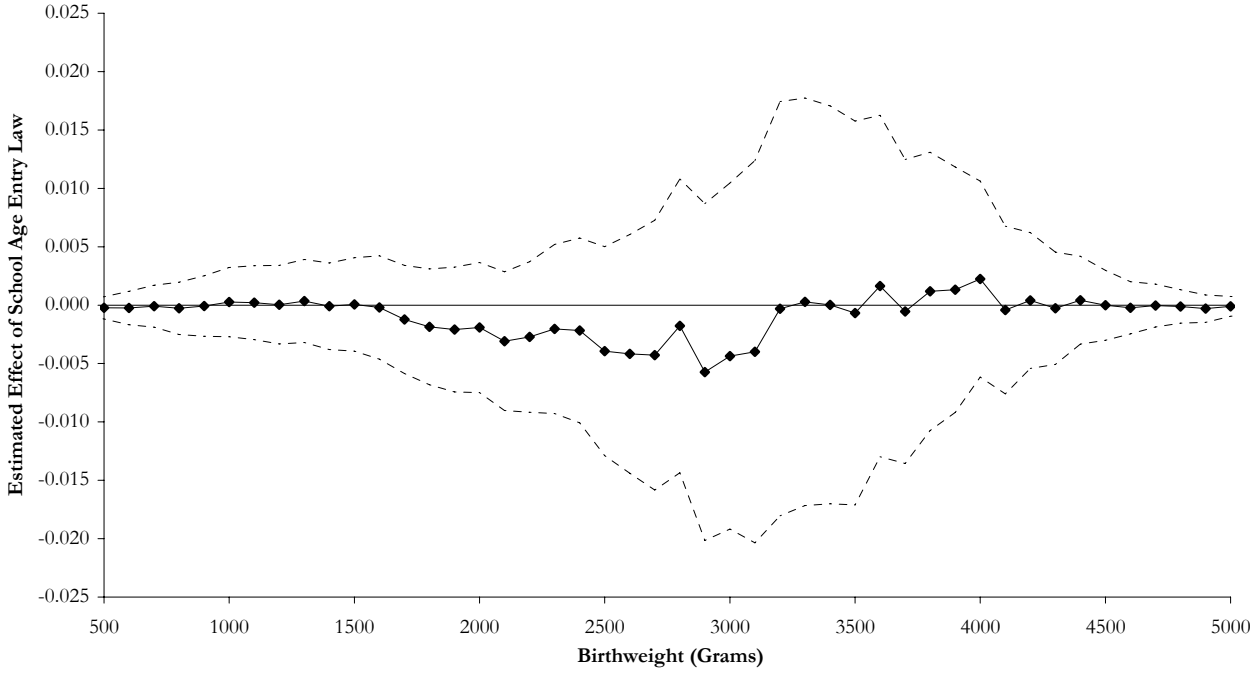
Notes: The open circles represent conditional means of low birth weight by mother's day of birth (relative to the school age entry cutoff date). The controls are fixed effects (uninteracted) for mother's ethnicity/race, mother's age, and year of infant's birth. The solid lines represent an estimate of a third degree polynomial which approximates the relationship between day of birth and low birth weight. See equation (9). The estimation sample includes only first-time mothers born in the state in which they gave birth who are 23 years old or younger and had singleton births.

Figure 9. Distribution of Estimated Birthweight Effects

A. California



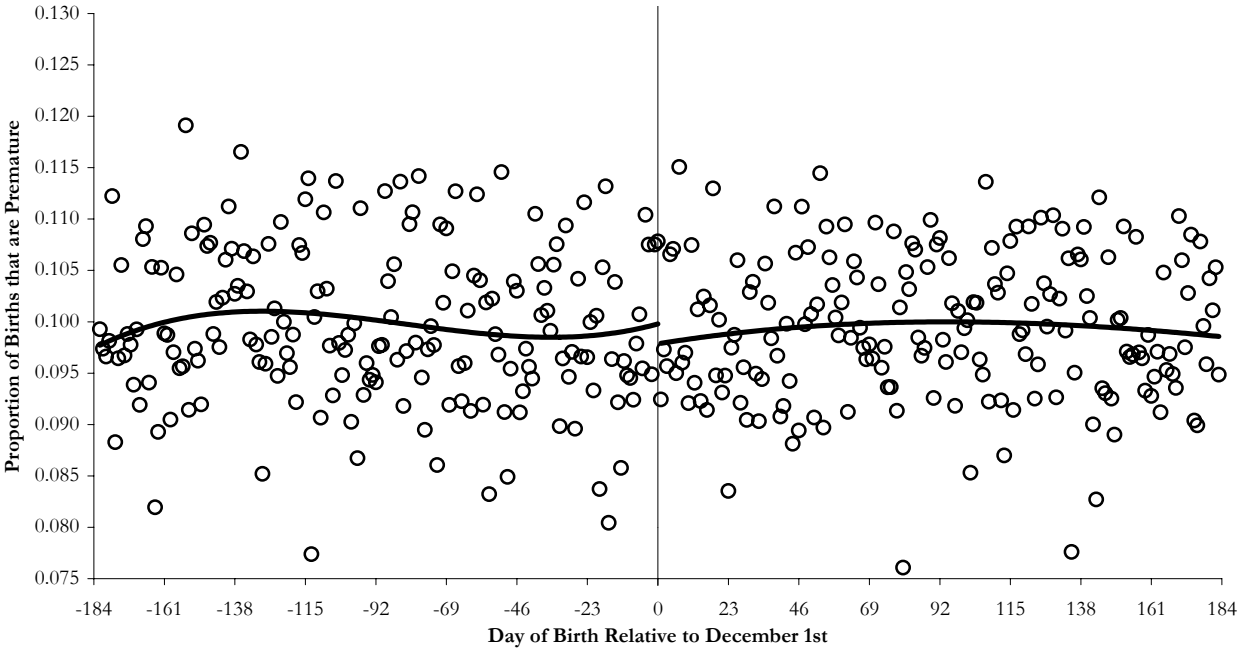
B. Texas



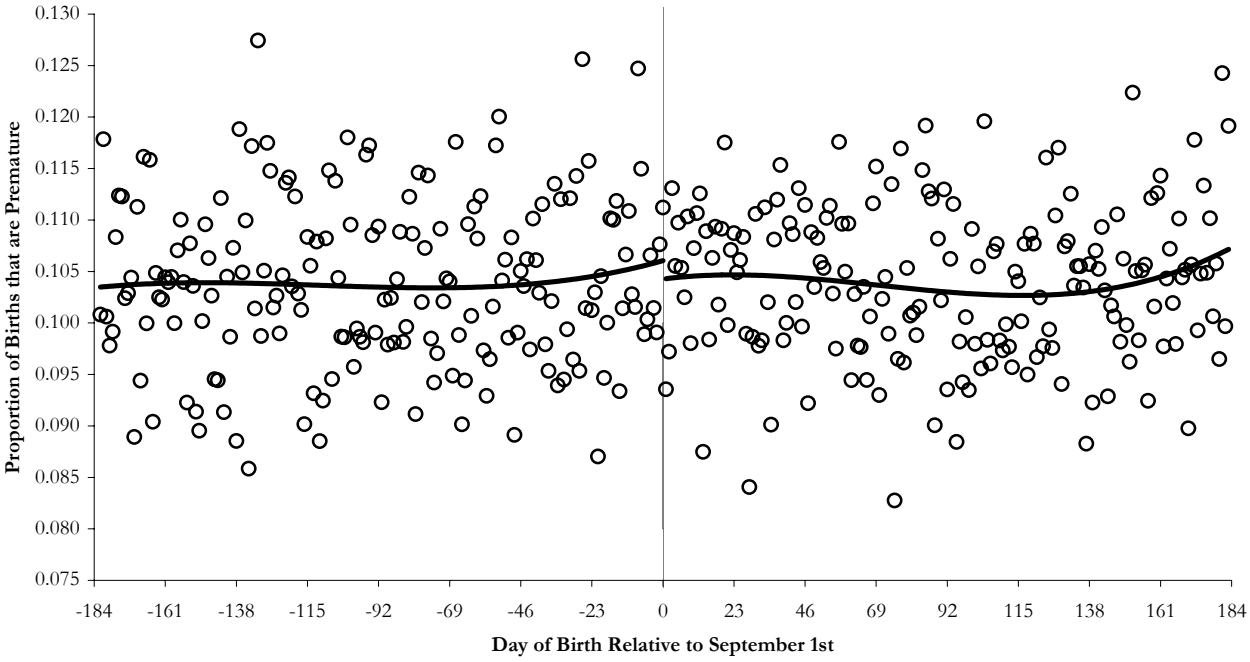
Notes: For details on the estimation sample, see Figure 7. Each point on the solid line represents the estimated difference in the probability of a baby born with a birth weight less than or equal to the weight represented on the horizontal axis, between the sample of mothers born before and after the school start-age entry date. This difference in probability comes from an estimate of equation (12) with birth weight replacing schooling as the dependent variable. The dashed lines represent the upper and lower 95% uniform confidence interval bands for the estimates represented by the solid lines. For more details on the construction of the uniform confidence intervals, see Appendix II.

Figure 10. Incidence of Prematurity, by Mother's Day of Birth: Native First-Time Mothers 23 Years Old and Younger

A. California



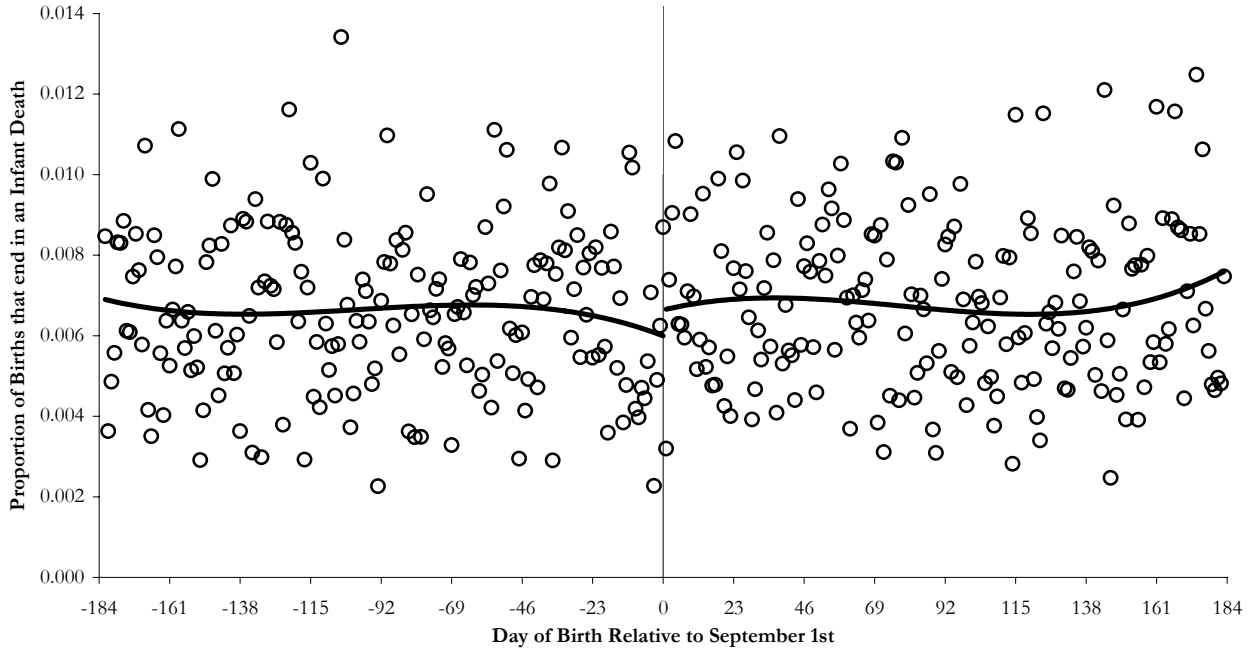
B. Texas



Notes: The open circles represent conditional means of prematurity by mother's day of birth (relative to the school age entry cutoff date). The controls are fixed effects (uninteracted) for mother's ethnicity/race, mother's age, and year of infant's birth. The solid lines represent an estimate of a third degree polynomial which approximates the relationship between day of birth and prematurity. See equation (9). The estimation sample includes only first-time mothers born in the state in which they gave birth who are 23 years old or younger and had singleton births.

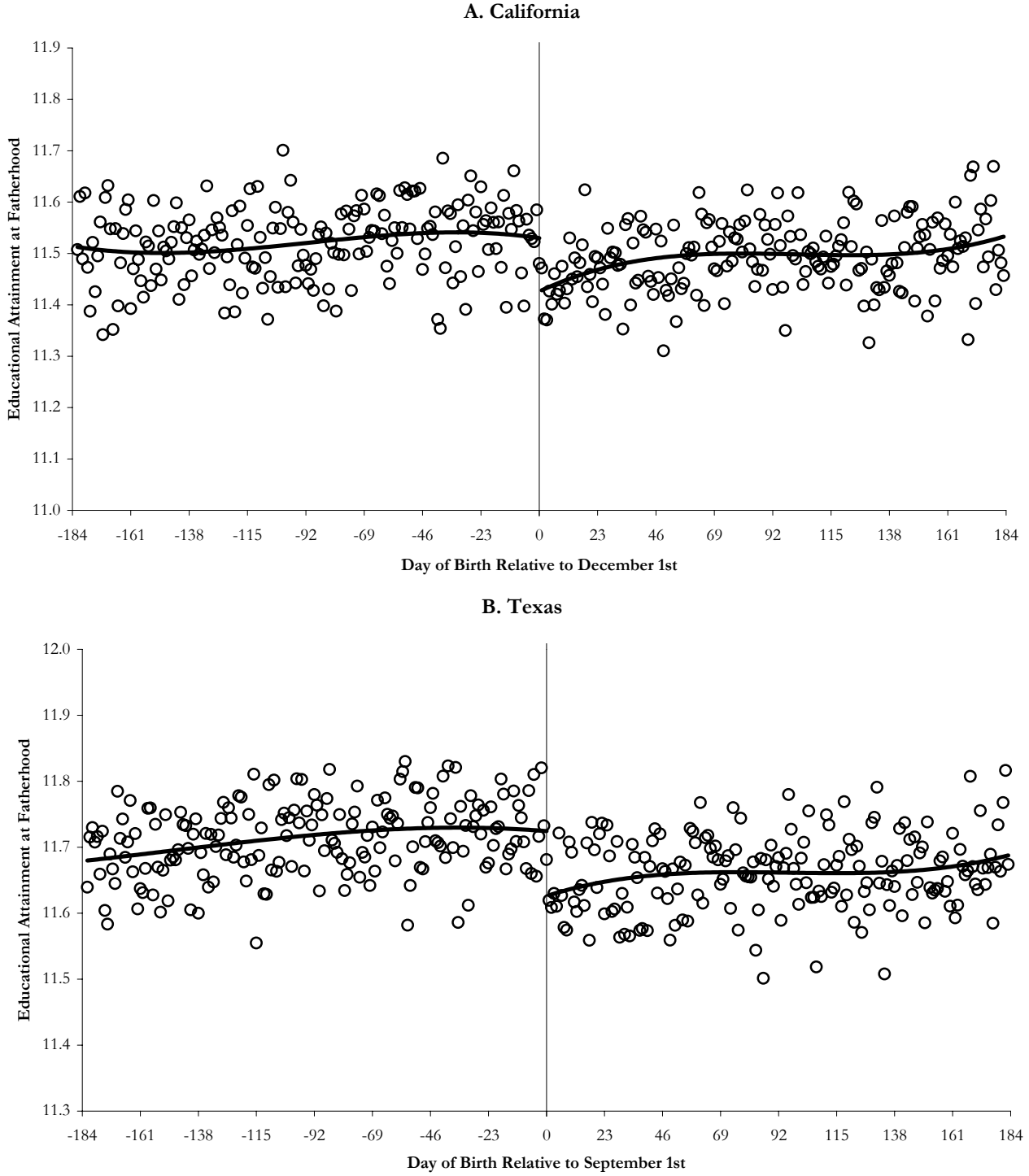
Figure 11. Incidence of Infant Death, by Mother's Day of Birth: Native First-Time Mothers 23 Years Old and Younger

A. Texas



Notes: The open circles represent conditional means of infant mortality by mother's day of birth (relative to the school age entry cutoff date). The controls are fixed effects (uninteracted) for mother's ethnicity/race, mother's age, and year of infant's birth. The solid lines represent an estimate of a third degree polynomial which approximates the relationship between day of birth and infant mortality. See equation (9). The estimation sample includes only first-time mothers born in the state in which they gave birth who are 23 years old or younger and had singleton births.

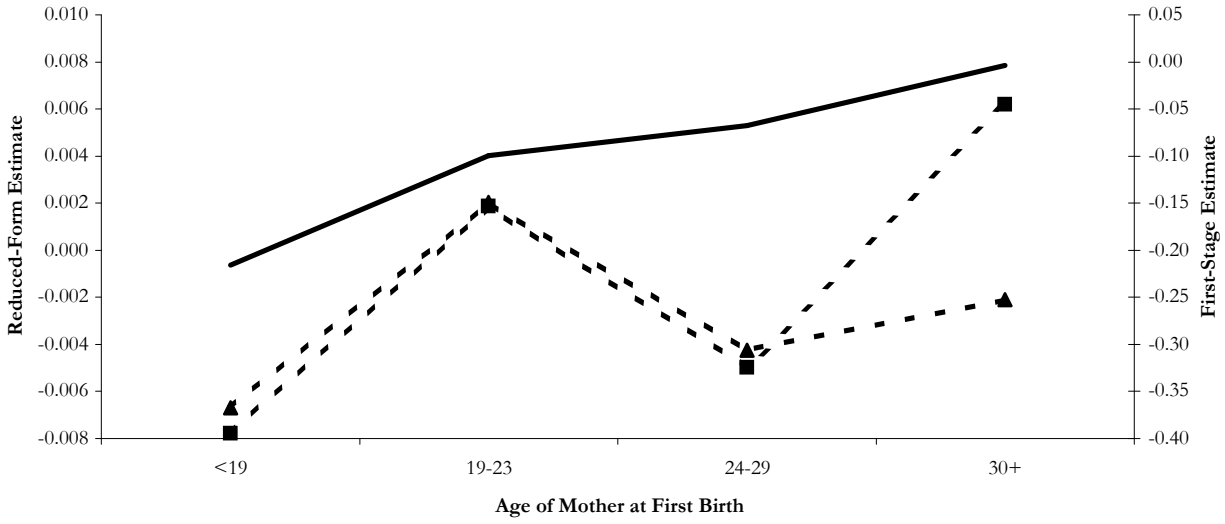
Figure 12. Paternal Education, by Mother's Day of Birth: Native First-Time Mothers 23 Years Old and Younger



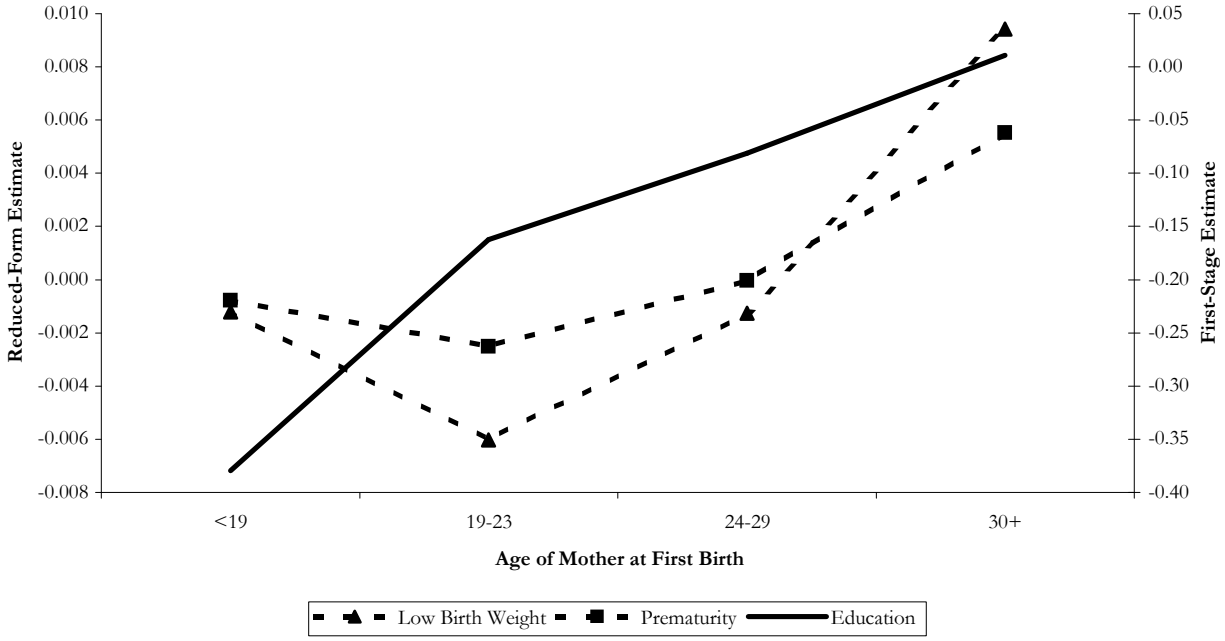
Notes: The open circles represent conditional means of father's education by mother's day of birth (relative to the school age entry cutoff date). The controls are fixed effects (uninteracted) for mother's ethnicity/race, mother's age, and year of infant's birth. The solid lines represent an estimate of a third degree polynomial which approximates the relationship between day of birth and father's education. See equation (9). The estimation sample includes only first-time mothers born in the state in which they gave birth who are 23 years old or younger and had singleton births.

Figure 13. First-Stage and Reduced-Form Estimates by Maternal Age
Native First-Time Mothers

A. California



B. Texas



Notes: The left y-axis values represent the reduced-form estimate of the school entry policies on either low birth weight or prematurity. The right y-axis values represent the first-stage estimate of the school entry policies on maternal education.

Table 1. Descriptive Statistics

	California (1989-2002)			Texas (1989-2001)		
	All Mothers	First-Time Mothers	First-Time Native Mothers Under 24	All Mothers	First-Time Mothers	First-Time Native Mothers Under 24
% Mothers White Non-Hispanic	36.09	39.15	38.71	42.96	45.22	39.78
% Mothers White Hispanic	45.25	41.26	43.64	41.90	39.47	40.66
%Mothers Black	7.19	6.81	13.88	12.32	12.12	19.25
%Mothers Asian	10.70	12.07	2.49	2.56	2.95	0.14
%Mothers Other	0.79	0.72	1.29	0.27	0.26	0.20
Mother's education (years)	11.83	12.37	11.64	11.93	12.19	11.26
	[3.51]	[3.2]	[1.69]	[3.06]	[2.93]	[1.86]
Mother's age (years)	27.19	24.79	19.17	25.86	23.44	19.01
	[6.17]	[6.09]	[2.31]	[5.98]	[5.66]	[2.32]
% Low birth weight (<2500 grams)	4.92	5.82	6.15	5.90	7.01	7.88
% Premature (<37 weeks of gestation)	9.25	9.16	9.95	9.00	9.13	10.39
Infant mortality rate (deaths before 1 year/1000 births)	NA	NA	NA	5.71	5.70	6.70
% Mothers smoking during pregnancy	2.03	1.58	2.90	7.80	6.33	8.48
% Mothers drinking during pregnancy	NA	NA	NA	1.18	1.03	0.97
% Mothers with sexually-transmitted diseases	1.34	1.52	1.68	2.24	2.60	3.81
% Mothers with prenatal care	98.84	99.06	99.06	97.37	97.83	98.02
% Prenatal care began in the 1st trimester	79.04	81.06	74.17	75.01	76.60	69.14
Number of prenatal care visits	11.39	11.75	11.41	10.91	11.21	10.80
	[4.13]	[4.08]	[4.08]	[4.51]	[4.46]	[4.36]
% Father present	90.97	89.42	84.14	84.03	80.44	69.27
Father's age (years)	30.00	27.85	22.16	29.05	26.81	22.16
	[6.97]	[6.94]	[4.18]	[6.71]	[6.5]	[4.18]
Father's education (years)	11.89	12.30	11.51	12.32	12.60	11.69
	[3.82]	[3.66]	[2.72]	[3.21]	[2.99]	[1.96]
% Having first birth	39.57	100.00	100.00	40.91	100.00	100.00
Observations	7,515,248	2,973,566	697,572	4,003,275	1,637,607	564,217

Notes: Mothers with missing education, parity, or birth date values or non-singleton births are excluded. The native subsample includes only mothers born in that state. Father's presence is measured by the presence of his educational attainment and birthdate on the birth certificate. Standard deviations are in brackets.

Table 2. The Effect of School Age Entry Laws: First-Stage and Reduced-Form Estimates

California			Texas		
Maternal Characteristics			**Maternal Characteristics**		
Mother's Education	Probability of Motherhood	Mother's Age	Mother's Education	Probability of Motherhood	Mother's Age
-0.1449	-0.0015	0.0177	-0.2559	-0.0029	0.0555
(0.0169)	(0.0031)	(0.0406)	(0.0123)	(0.0039)	(0.0490)
Birth Outcomes			**Birth Outcomes**		
Low birthweight	Premature	Infant Death	Low birthweight	Premature	Infant Death
-0.0014	-0.0019	NA	-0.0039	-0.0018	0.0006
(0.0022)	(0.0030)		(0.0024)	(0.0030)	(0.0009)
Risky Maternal Behaviors			**Risky Maternal Behaviors**		
Mother Smokes	Mother Drinks	STDs	Mother Smokes	Mother Drinks	STDs
0.0025	NA	0.0002	-0.0002	-0.0021	-0.0013
(0.0013)		(0.0011)	(0.0028)	(0.0009)	(0.0022)
Prenatal Care			**Prenatal Care**		
Any Care	Care in First Trimester	Number of Visits	Any Care	Care in First Trimester	Number of Visits
-0.0003	-0.0024	-0.0283	0.0007	-0.0108	-0.0592
(0.0010)	(0.0049)	(0.0384)	(0.0012)	(0.0043)	(0.0472)
Paternal Characteristics			**Paternal Characteristics**		
Father Present	Father's Age	Father's Education	Father Present	Father's Age	Father's Education
0.0012	-0.1109	-0.1039	-0.0004	-0.2334	-0.0983
(0.0038)	(0.0333)	(0.0237)	(0.0045)	(0.0495)	(0.0213)

Notes: The table gives estimated discontinuities in the specified outcome in mother's birthday at the school entry date. The estimation sample includes only first-time mothers born in the state in which they gave birth who are 23 years old or younger and had singleton births, with the exception of the maternal age regressions, where we do not exclude mothers 24 and older from the sample. Each estimate stems from a global polynomial regression of the specified outcome on a cubic polynomial in mother's birthday (normalized to zero at the school entry date), a dummy for whether the mother was born after the school entry date, and an interaction of a cubic polynomial with the post-cutoff dummy. The discontinuity estimate is the coefficient on the post-cutoff dummy. In addition to the functional form for mother's birthday just described, each regression additionally includes controls for the mother's ethnicity/race, the mother's age (except for the maternal age regressions), and the year of the infant's birth. Standard errors, presented in parentheses, are clustered standard errors at the level of mother's birthday.

Table 3. Instrumental Variable Estimates of the Effect of Maternal Education on Infant Health

	California			Texas		
	Low birthweight	Premature	Infant death	Low birthweight	Premature	Infant death
IV/RD	0.0096	0.0134	NA	0.0154	0.0070	-0.0025
(Just-identified)	(0.0153)	(0.0210)		(0.0098)	(0.0121)	(0.0035)
IV/DOB	0.0006	-0.0001	NA	0.0099	0.0011	-0.0011
(Over-identified)	(0.0044)	(0.0062)		(0.0044)	(0.0052)	(0.0013)
J Test	3.38	4.20	NA	5.93	4.06	3.00
	[0.76]	[0.65]		[0.43]	[0.67]	[0.81]
Hausman Test of	0.37	0.47	NA	0.39	0.29	0.19
IV/RD - IV/DOB	[0.54]	[0.49]		[0.53]	[0.59]	[0.66]
Difference						

Notes: For details on the estimation sample, control variables, and regression specification, see notes to Table 2. The presented coefficients represent the effect of an additional year of maternal education on the specified outcome variable. Estimates labelled "IV/RD" use an indicator for whether the mother's birthday is subsequent to the school entry date as the excluded instrument for maternal education and use as included instruments a cubic in mother's birthday and the interaction of a cubic in mother's birthday with the post-cutoff dummy. Estimates labelled "IV/DOB" use all of the aforementioned variables as excluded instruments. Standard errors, in parentheses, are clustered standard errors at the level of mother's birthday. The row labelled "J Test" gives an omnibus test of overidentification (see text for details). The test is distributed chi-square with six degrees of freedom under the composite null hypothesis of no misspecification. The row labelled "Hausman Test" is a test of the difference between the regression discontinuity and two-stage least squares estimates (see text for details). This test is distributed F with one numerator degree of freedom and a large denominator degree of freedom under the null hypothesis that the two-stage least squares and regression discontinuity estimates share a probability limit. P-values for both tests are given in brackets.

Table 4. Pooled Instrumental Variable Estimates of the Effect of Maternal Education on Infant Health and Health Inputs

	Infant Health		
	Low Birthweight	Premature	Infant Death
IV/RD	0.0122	0.0105	NA
	(0.0095)	(0.0127)	
IV/DOB	0.0048	0.0004	NA
	(0.0031)	(0.0041)	
	Risky Maternal Behaviors		
	Mother Smokes	Mother Drinks	Mother Has an STD
IV/RD	-0.0100	NA	0.0012
	(0.0070)		(0.0058)
IV/DOB	0.0024	NA	0.0009
	(0.0025)		(0.0020)
	Prenatal Care		
	Any Care	Care in First Trimester	Number of Visits
IV/RD	-0.00003	0.0280	0.2160
	(0.0042)	(0.0196)	(0.1708)
IV/DOB	0.0011	0.0307	0.2041
	(0.0015)	(0.0056)	(0.0541)
	Father's Characteristics		
	Father Present	Father's Age	Father's Education
IV/RD	-0.0040	0.8376	0.5780
	(0.0163)	(0.1789)	(0.0961)
IV/DOB	0.0106	0.8249	0.3235
	(0.0050)	(0.0578)	(0.0359)

Notes: These estimates are weighted averages of the estimates in Tables 3 and 5. The weights are the number of observations underlying the parameter estimates. Standard errors are presented in parentheses.

Table 5. Instrumental Variable Estimates of the Effect of Maternal Education on Risk Factors for Poor Infant Health

	California			Texas		
	Risky Maternal Behaviors			**Risky Maternal Behaviors**		
	Mother Smokes	Mother Drinks	Mother Has an STD	Mother Smokes	Mother Drinks	Mother Has an STD
IV/RD	-0.0173	NA	-0.0016	0.0006	0.0080	0.0050
	(0.0093)		(0.0080)	(0.0107)	(0.0035)	(0.0083)
IV/DOB	0.0042	NA	-0.0009	-0.0001	0.0021	0.0032
	(0.0029)		(0.0023)	(0.0046)	(0.0016)	(0.0034)
	Prenatal Care			**Prenatal Care**		
	Any Care	Care in First Trimester	Number of Visits	Any Care	Care in First Trimester	Number of Visits
IV/RD	0.0021	0.0163	0.1963	-0.0027	0.0425	0.2410
	(0.0066)	(0.0325)	(0.2658)	(0.0047)	(0.0172)	(0.1912)
IV/DOB	0.0012	0.0333	0.2453	0.0010	0.0274	0.1519
	(0.0020)	(0.0082)	(0.0753)	(0.0022)	(0.0075)	(0.0769)
	Father's Characteristics			**Father's Characteristics**		
	Father Present	Father's Age	Father's Education	Father Present	Father's Age	Father's Education
IV/RD	-0.0084	0.7778	0.6951	0.0014	0.9307	0.3901
	(0.0259)	(0.2652)	(0.1478)	(0.0174)	(0.1969)	(0.0797)
IV/DOB	0.0023	0.9226	0.2914	0.0208	0.6730	0.3750
	(0.0070)	(0.0795)	(0.0541)	(0.0072)	(0.0810)	(0.0347)

Notes: For details on the estimation sample, control variables, and regression specification, see notes to Table 2. The presented coefficients represent the effect of an additional year of maternal education on the specified outcome variable. See notes to Table 3 for a description of the two estimators. Standard errors, in parentheses, are clustered standard errors at the level of mother's birthday.

Table 6. Instrumental Variable Estimates of the Effect of Total Parental Education on Infant Health and Health Inputs

	California			Texas		
	Infant Health			**Infant Health**		
	Low Birthweight	Premature	Infant Death	Low Birthweight	Premature	Infant Death
IV/RD	0.0045	0.0063	NA	0.0204	0.0083	0.0001
	(0.0086)	(0.0135)		(0.0095)	(0.0089)	(0.0029)
IV/DOB	-0.0002	-0.0027	NA	0.0081	0.0013	-0.0007
	(0.0036)	(0.0051)		(0.0039)	(0.0042)	(0.0011)
	Risky Maternal Behaviors			**Risky Maternal Behaviors**		
	Mother Smokes	Mother Drinks	Mother Has an STD	Mother Smokes	Mother Drinks	Mother Has an STD
IV/RD	-0.0103	NA	-0.0011	0.0013	0.0069	0.0027
	(0.0055)		(0.0046)	(0.0086)	(0.0026)	(0.0065)
IV/DOB	0.0026	NA	-0.001	-0.0030	0.0014	0.002
	(0.0023)		(0.0019)	(0.0038)	(0.0013)	(0.0027)
	Prenatal Care			**Prenatal Care**		
	Any Care	Care in First Trimester	Number of Visits	Any Care	Care in First Trimester	Number of Visits
IV/RD	0.0013	0.0073	0.0654	0.0008	0.0336	0.2355
	(0.0032)	(0.0185)	(0.1621)	(0.0038)	(0.0153)	(0.1849)
IV/DOB	0.0005	0.0272	0.1937	0.0009	0.0181	0.0879
	(0.0014)	(0.0066)	(0.0588)	(0.0017)	(0.0063)	(0.0687)

Notes: For details on the estimation sample, control variables, and regression specification, see notes to Table 2. The presented coefficients represent the effect of an additional year of total parental education (sum of maternal and paternal education) on the specified outcome variable. Estimates labelled "IV/RD" use an indicator for whether the mother's birthday is subsequent to the school entry date as the excluded instrument for parental education and use as included instruments a cubic in mother's birthday and the interaction of a cubic in mother's birthday with the post-cutoff dummy. Estimates labelled "IV/DOB" use all of the aforementioned variables as excluded instruments. Standard errors, in parentheses, are clustered standard errors at the level of mother's birthday.

Table 7. Tests of Overidentification: Smoothness and Flatness in Pre-determined Characteristics

	California					Texas				
	Maternal Characteristics					**Maternal Characteristics**				
	Hispanic	Black	Black*	Low Birthweight*	High Birthweight*	Hispanic	Black	Black*	Low Birthweight*	High Birthweight*
Discontinuity	-0.0078	-0.0066	0.0004	-0.0011	0.0013	0.0052	-0.0010	0.0011	0.0000	0.0000
	(0.0054)	(0.0037)	(0.0012)	(0.0009)	(0.0012)	(0.0054)	(0.0047)	(0.0016)	(0.0015)	(0.0010)
Test of Identifying Assumption of IV/RD Estimators	2.08	3.29	0.08	1.49	1.25	0.93	0.05	0.44	0.00	0.00
	[0.150]	[0.071]	[0.774]	[0.223]	[0.264]	[0.336]	[0.832]	[0.506]	[0.982]	[0.975]
Test of Identifying Assumptions of IV/DOB Estimator	3.76	5.89	9.60	17.63	44.22	20.25	3.70	70.83	42.30	49.97
	[0.001]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.001]	[0.000]	[0.000]	[0.000]
	Grandmaternal Characteristics*					**Grandmaternal Characteristics***				
	Black	Age at Childbirth	Native Born	Parity	Survival Rate of Children	Black	Age at Childbirth	Native Born	Parity	Survival Rate of Children
Discontinuity	0.0000	0.0534	0.0033	0.0056	0.0008	0.0016	-0.0350	0.0014	0.0079	0.0023
	(0.0012)	(0.0274)	(0.0027)	(0.0058)	(0.0014)	(0.0017)	(0.0248)	(0.0027)	(0.0074)	(0.0013)
Test of Identifying Assumption of IV/RD Estimators	0.00	3.79	1.45	0.95	0.35	0.89	2.00	0.25	1.15	3.42
	[0.991]	[0.052]	[0.229]	[0.331]	[0.555]	[0.345]	[0.158]	[0.615]	[0.284]	[0.065]
Test of Identifying Assumptions of IV/DOB Estimator	9.24	27.28	75.98	12.52	13.65	75.06	27.38	10.44	13.73	20.29
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
	Grandpaternal Characteristics*					**Grandpaternal Characteristics***				
		Age at Delivery					Age at Delivery			
Discontinuity		0.0526					-0.0630			
		(0.0324)					(0.0303)			
Test of Identifying Assumption of IV/RD Estimators		2.64					4.33			
		[0.105]					[0.038]			
Test of Identifying Assumptions of IV/DOB Estimators		13.94					13.89			
		[0.000]					[0.000]			

Notes: The table presents estimated discontinuities in pre-determined maternal, grandmaternal, and grandpaternal characteristics using our administrative data from California and Texas, 1989-present, and the 1969-1988 public-use Natality Detail Files. Estimates based on the latter are indicated by an asterisk; other estimates are based on the former. Standard errors, given in parentheses, are clustered standard errors at the level of mother's birthday. The discontinuity estimates are the coefficient on the post-cutoff dummy from a regression of the outcome on a cubic in mother's birthday, a dummy for whether the mother was born after the school entry date, and a cubic in mother's birthday interacted with the post-cutoff dummy. The row labelled "test of identifying assumption of IV/RD estimator" gives the F-statistic for a single zero restriction on the coefficient on the post-cutoff dummy. These are distributed F with one numerator degree of freedom and a large denominator degree of freedom under the null hypothesis of continuity of confounders in mother's birthday. The row labelled "test of identifying assumptions of IV/DOB estimator" gives the F-statistic for a joint zero restriction on all variables in the global polynomial model described. These are distributed F with seven numerator degrees of freedom and a large denominator degree of freedom under the null hypothesis of no seasonality of confounders in mother's birthday.

Table 8. Test of Age-for-Grade Effects in the Probability of Motherhood

State	Date Change	Post-Change	Mean Probability of Motherhood
Florida	1/1-9/1	0.0052 (0.0045)	0.3724
Illinois	12/1-9/1	0.0018 (0.0016)	0.0631
Mississippi	12/1-9/1	0.0066 (0.0096)	0.5218
Vermont	1/1-8/31	0.0037 (0.0143)	0.1987
Wisconsin	12/1-9/1	-0.0004 (0.0055)	0.3154

Notes: The second column in this table gives the change in the school entry cutoff date. The third column presents the estimated post-cutoff-change dummy coefficient along with the standard error of this estimate in parentheses. These coefficient estimates come from regressions of the probability of motherhood by date of birth on day of birth fixed effects, a linear cohort trend, and a post cutoff change dummy. The estimation sample include all females born up to 3 years preceeding or up to 3 years following the final date change. All fractions have been trimmed such that each cohort is observed for the same ages. The fourth column in this table is the mean fraction of women in the estimation sample observed giving birth.

Table 9. Reduced-Form and Instrumental Variables Estimates of the Effect of Education on Delivery Payment (CA) and Location of Prenatal Care (TX)

	Reduced Form Estimates					
	California			Texas		
	Public Payment for Delivery	Private Payment for Delivery	Self-Payment for Delivery	Prenatal Care Received at Hospital	Prenatal Care Received at Public Clinic	Prenatal Care Received at Private Clinic
Dependent Variable Mean	0.5772	0.3955	0.0252	0.1643	0.1590	0.6888
Discontinuity Estimate	0.0102	-0.0081	-0.0020	0.0093	0.0015	-0.0088
	(0.0044)	(0.0039)	(0.0015)	(0.0045)	(0.0045)	(0.0047)
	Instrumental Variables Estimates					
	California			**Texas**		
IV/RD	-0.0712	0.0565	0.0138	-0.0350	-0.0055	0.0332
	(0.0296)	(0.0255)	(0.0108)	(0.0171)	(0.0169)	(0.0175)
IV/DOB	-0.0604	0.0586	0.0015	-0.0106	-0.0204	0.0287
	(0.0090)	(0.0084)	(0.0032)	(0.0075)	(0.0074)	(0.0093)

Notes: For notes on the estimation sample, control variables, and regression specification for the reduced-form estimates, see Table 2. For notes on the estimation sample, control variables, and regression specification for the instrumental variables estimates, see Table 3. A mother can report more than one location for prenatal care. The prenatal care regressions are not conditional on having received care.

Table A1. The Effect of School-Age Entry Laws: Probability of the Outcome Variable Missing

California			Texas		
Birth Outcomes			**Birth Outcomes**		
Low birthweight	Premature	Infant death	Low birthweight	Premature	Infant death
-0.0001	0.0003	NA	0.0004	-0.0007	NA
(0.0001)	(0.0017)		(0.0003)	(0.0005)	
Risky Maternal Behaviors			**Risky Maternal Behaviors**		
Mother Smokes	Mother Drinks	STDs	Mother Smokes	Mother Drinks	STDs
-0.0002	NA	0.0060	-0.0016	-0.0019	-0.0034
(0.0002)		(0.0026)	(0.0035)	(0.0035)	(0.0036)
Prenatal Care			**Prenatal Care**		
Any Care	Care in First Trimester	Number of Visits	Any Care	Care in First Trimester	Number of Visits
0.0011	0.0021	0.0015	0.0026	0.0022	0.0034
(0.0007)	(0.0009)	(0.0014)	(0.0014)	(0.0014)	(0.0019)
Father's Characteristics			**Father's Characteristics**		
Father Present	Father's Age	Father's Education	Father Present	Father's Age	Father's Education
NA	0.0000	-0.0035	NA	-0.0046	-0.0033
	(0.0035)	(0.0037)		(0.0050)	(0.0051)

Notes: For notes on the estimation sample, control variables, and regression specification, see Table 2. The table gives discontinuity estimates for the probability of the specified outcomes being missing. Each coefficient stems from a regression of an indicator for missing for the specified outcome on a cubic in mother's birthday, a dummy for whether the mother was born after the school entry date, and a cubic in mother's birthday interacted with the post-cutoff dummy. Standard errors, given in parentheses, are clustered standard errors at the level of mother's birthday. Father's presence is available for all births by definition.

Table A2. Robustness of Estimates in Table 2

California			Texas		
Maternal Characteristics			**Maternal Characteristics**		
Mother's Education	Probability of Motherhood	Mother's Age	Mother's Education	Probability of Motherhood	Mother's Age
1.06	1.42	1.06	1.05	1.21	1.14
[0.227]	[0.000]	[0.209]	[0.264]	[0.004]	[0.034]
Birth Outcomes			**Birth Outcomes**		
Low birthweight	Premature	Infant Death	Low birthweight	Premature	Infant Death
1.03	1.13	NA	0.92	1.02	1.00
[0.361]	[0.047]		[0.854]	[0.386]	[0.467]
Risky Maternal Behaviors			**Risky Maternal Behaviors**		
Mother Smokes	Mother Drinks	STDs	Mother Smokes	Mother Drinks	STDs
1.03	NA	0.92	0.89	0.93	1.03
[0.359]		[0.862]	[0.940]	[0.810]	[0.338]
Prenatal Care			**Prenatal Care**		
Any Care	Care in First Trimester	Number of Visits	Any Care	Care in First Trimester	Number of Visits
1.17	1.04	1.00	1.02	1.06	1.20
[0.013]	[0.308]	[0.513]	[0.388]	[0.205]	[0.006]
Paternal Characteristics			**Paternal Characteristics**		
Father Present	Father's Age	Father's Education	Father Present	Father's Age	Father's Education
1.00	1.10	1.18	1.10	1.11	1.05
[0.470]	[0.090]	[0.010]	[0.098]	[0.074]	[0.263]

Notes: The table reports tests of the robustness of the global polynomial approximation to the conditional expectation of the specified outcome in mother's birthday that was used in estimating the discontinuities presented in Table 2. Each entry in the table (excepting probability of motherhood) is an F-statistic that compares the fit of a completely saturated model in mother's birthday (i.e., 366 indicators for each possible birthday) to the more parsimonious model underlying the estimates in Table 2: a cubic in mother's birthday, a dummy for whether the mother was born after the cutoff date, and a cubic in mother's birthday interacted with the post-cutoff dummy. The description of the entry for the probability of motherhood is identical, but a seventh order polynomial replaces the cubic. The numerator degrees of freedom for the F-statistic is 358, except for the probability of motherhood, where the degrees of freedom is 350, and the denominator degrees of freedom is large. P-values are given in brackets.